

BACKGROUND STUDY :

Just like water, electricity/power, education and health, food are considered essential public services, the role of urban green spaces is equally important in establishing and maintaining quality of life in a community. They contribute to health and environmental wellbeing of a community and the region. The physical elements and activities in the urban green spaces offer many benefits to the citizens, namely health, social interaction and economic value. They provide identity and sense of place to the citizens.

Urbanization results in an increasing proportion of the population living in cities. Urban living limits access to nature and can increase exposure to certain environmental hazards, such as air and noise pollution. Many urban areas face increasing pressure from expanding populations, limited resources and growing impacts of climate change. These challenges must be addressed in order for cities to provide healthy and sustainable living environments.

The green spaces are the most significant social spaces for people to interact on a daily basis and also considered as one of the sustainability indicators for maintaining the well-being in residential Communities. The benefits of green space for wellbeing are extensively recognized and progressively more documented. Due to increasing urbanization and housing demand, Residential communities are growing in suburbs and few in the urban core. Due to which depletion in per capita green spaces are recorded. Much of this work focus on the potential of natural environments to provide opportunities to enhance wellbeing in different ways:

- Physical wellbeing through physical activity and fresh air.
- Mental wellbeing through stress reduction and attention restoration.
- Social wellbeing through social integration, engagement and participation.

AIM:

To safe guard the future of green spaces in residential areas and to improve the green cover in residential neighbourhoods.

OBEJECTIVES:

- To study and understand the importance of green spaces in an unplanned residential area.
- To make urban areas more attractive by controlling and minimizing environmental hazards such as air pollution or noise are reduced
- Highlight the importance of green spaces in urban areas and the values within.
- To develop the urban parks to improve environmental conditions provide healthy urban conditions for good physical and mental well-being.

SCOPE AND LIMITATIONS:

The scope of the study focuses on increasing of green cover in residential areas in order to improve environmental conditions, where there is lack of green spaces due to increase in urbanisation which effects people with health conditions & increase in pollution.

EXPECTED OUTCOMES:

To implement improving the green spaces in an residential areas. Improving the and planning accordingly for and healthy environmental urban sprawl.



METHODOLOGY

INTRODUCTION – BACKGROUND STUDY

AIM

To safe guard the future of green spaces in residential areas and to improve the quality of urban areas especially neighbourhood areas and recreational areas by increasing the green environment.

OBJECTIVES:

To study and understand the importance of green spaces in an unplanned residential area.

To make urban areas more attractive by controlling air, water pollution and by minimizing sound pollution.

Implementation of alternative approaches like (vertical landscape gardens, green walls, green roofs etc.) in residential neighbourhood .

Defining Green Growth and its potential in the Indian Cities context

LITERATURE REVIEW

Concept of urban green spaces, its role , importance , benefits,

Studying the existing scenario

International and national case studies.

BEST PRACTICES

STUDY AREA SAIDABAD, HYD.

DATA COLLECTION AND COMPIlation

DATA ANALYSIS

RECOMMENDATION

PRIMARY DATA:

Landcover data, existing scenario, land use, survey.

SECONDARY DATA:

Guidelines, regulations.



LITERATURE REVIEW

INTRODUCTION :



Since the beginning of Man kind, nature has provided humans with shelter, resources, etc., Being an important part of their development. Although this relationship has changed and evolved from one of the main aspect in human life to areas dedicated to recreation and similar activities, the relationship between humans and nature is still present. This relationship is easily manifest in rural areas, where the population has a direct relationship with the nature, either by direct access to it, to the growth of food, etc.

But in urban areas, this relationship is more limited, and it usually is directly related to the presence of green spaces in the city. Moreover, these green spaces have traditionally been considered as 'good places' for people. By this their presence and importance in cities are understandable. There are multiple definitions of green spaces. They define green space as an "open, undeveloped land with natural vegetation", a definition that includes areas such as parks, forests, vegetation corridors, playing fields, wetlands, urban forests, cemeteries etc.

This thesis in regards of green spaces, the main focus has been put into to investigate if there are any benefits of green spaces, as well as in articles reflecting upon /urbanization processes and green spaces. The studies reflecting upon the relationship between green spaces and the benefits they have on human population has been increasing lately, as an answer to concern about issues produced by urbanization, in relation to the environment, and changes in human lifestyles. The relationship between the presence of green spaces and their positive effects was done through articles that have researched. Literature about green spaces and their benefits was wide, but was found somehow incomplete, as most of the articles related to a limited number of effects of these green spaces, but most of them fail to do a complete study.

Green spaces and other nature-based solutions offer innovative approaches to increase the quality of urban settings, enhance local resilience and promote sustainable lifestyles, improving both the health and the well-being of urban residents. Parks, playgrounds or vegetation in public and private places are a central component of these approaches and can help to ensure that:

- urban residents have adequate opportunities for exposure to nature;
- urban biodiversity is maintained and protected;
- environmental hazards such as air pollution or noise are reduced.
- The impacts of extreme weather events (heatwaves, extreme rainfall or flooding) are mitigated.
- The quality of urban living is enhanced.
- The health and well-being of residents is improved.

Such positive experiences affect their happiness level, which is an important component of well being thus leading to sustainable lives. The immediate ecological benefits of green spaces are reduction in the urban heat island effect, cleaner air, and reduced noise pollution, which have been shown to improve health-related quality of life, and reduce both morbidity and mortality.

CONCEPT OF URBAN GREEN SPACES:

The world's cities are becoming increasingly congested and polluted. Urban green spaces provide a wide range of ecosystem services that could help combat many urban ills and improve life for city dwellers.

The definition of urban green spaces which is agreed on by ecologists, economists, social scientists and planners: "they are public and private open spaces in urban areas, primarily covered by vegetation which is directly or indirectly available for users. In another way, it is the land that consists predominantly of unsealed, permeable, "soft" surfaces such as soil, grass, shrubs and trees. It is the umbrella term for all such areas whether or not they are publicly accessible or publicly managed. It includes all areas of parks, play areas and other green spaces specifically intended for recreational use.

Trees, grass and other green vegetation provide wildlife habitat, prevent soil erosion, absorb air pollutants, and lower surface temperatures. Lack of green spaces leads to higher air temperatures and more ground level ozone, with fewer trees and plants to clean the air and provide oxygen.

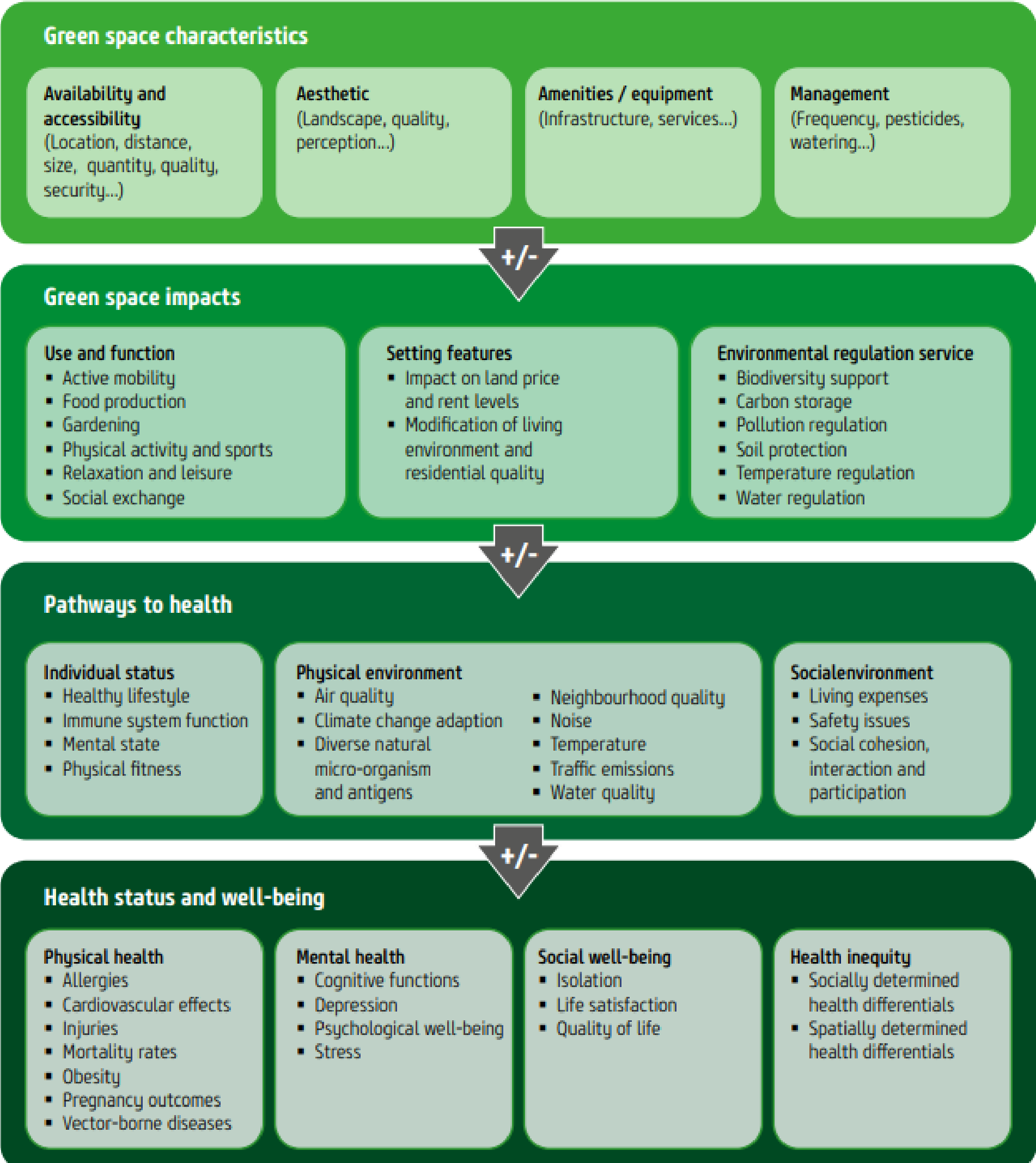
Source :Role of Green Spaces for Maintaining Well-Being in Residential Community Development By Prashanti Rao <https://www.intechopen.com/online-first/76802>



CHARACTERISTICS OF URBAN GREEN SPACES:

Such green spaces vary in size, vegetation cover, species richness, environmental quality, proximity to public transport, facilities and services. Public green spaces include parks and reserves, sporting fields, riparian areas like stream and river banks, greenways and trails, community gardens, street trees and nature conservation.

Public green spaces should be at the centre of neighbourhood and not more than five minutes' walk for most residents, public buildings or shops. Therefore, accessibility and proximity are very important factors to consider during planning and design of an urban green space.



In green space research, there is an increasing number of studies that concern people’s experience and use, but it has been pointed out that more research is needed due to increasing urbanization issues like safety and security. Many previous models and theories substantiate that the usability of green space is directly associated with its accessibility. The usability of green space differs due to different spatial arrangements within the residential communities and variation in the provision of it with respect to quantity and quality. Physical accessibility and the connection is a major research area in the field of green space use in the residential area because people are associated with the green space usage

WHAT DO GOOD URBAN GREEN SPACES OFFER:

Urban green spaces can give enormous benefits to the neighbourhood and community by improving health, social well-being and enhancing enjoyment of local environment. The important role of urban green spaces is recognized both in the character and the life they bring to towns.

IMPORTANCE OF GREEN ENVIRONMENTS FOR RESIDENTIAL COMMUNITIES:

In many countries like England, Australia, Germany and Belgium the concept of the green environment is popular they explored healthy city planning and defined it as an effective tool for achieving sustainable health and well-being in humans. It is important that these types of green environments are publicly accessible to provide spaces of social interaction for residents. The green spaces have diverse classifications, and each provides its own values and benefits. Thus, further exploration of green environments is critical to ensure higher accessibility, usability, and functionality of green environments for future residents. In many residential communities green spaces are buffers between Built and Exterior surroundings. In addition, Multi unit residential developments have focused on branding value and esthetic aspects and ignored the importance of natural spaces within the living environment.

However, the importance of green spaces in residential communities has been recognized by residents and in some communities they are developed and protected voluntary. Hence it is essential to understand the importance of green spaces for sustenance of the well being of residential communities and their overall development.

UNDERSTANDING TERMINOLOGIES

➤ Green spaces:

The presence of green spaces is considered as one of the important components for good quality of life in urban areas. For this reason, it is important to understand the scale, functions and benefits of green spaces ranging from regional to local scales/levels. In this context, a few definitions are discussed below. Green spaces is a broad terminology, which may be used for both macro level and micro level spatial scales. This is the definition of urban green spaces in the European context, which is concurred with among a majority of ecologists economists, social scientists and planners.

- Urban Greens are green spaces located in urban areas mainly covered by vegetation, which are directly used for active or passive recreation, or indirectly used by virtue of their positive influence on the urban environment, accessible to citizens, serving the diverse needs of citizens and thus enhancing the quality of life in cities or urban regions
- Green spaces are the ‘green lungs’ of our towns and cities which contribute to improving people’s physical and mental health by providing places for informal recreation – walking, cycling, sitting, socializing and children’s play – and ‘breathing spaces’ to take time out from the stresses of modern life. They include not only areas to which the public have physical access, but also visual access, for example, in the way green spaces provide setting for building, communities, and everyday activities.

Well being:

Well-being is the experience of health, happiness, and prosperity. It includes having good mental health, high life satisfaction, a sense of meaning or purpose, and ability to manage stress. More generally, well-being is just feeling well. Well-being is a positive outcome that is meaningful for people and for many sectors of society, because it tells us that people perceive that their lives are going well. Good living conditions (e.g., housing, employment) are fundamental to well-being. Wellbeing is important because it effects both physically and mentally, and is the essence of living. Researchers from different disciplines have examined different aspects of well-being that include the following

- Physical well-being.
- Economic well-being.
- Social well-being.
- Development and activity.
- Emotional well-being.
- Psychological well-being.
- Life satisfaction.
- Domain specific satisfaction.
- Engaging activities and work.



➤ GREEN SPACE FUNCTIONS AND BENEFITS:
General benefits:

The functions and benefits of green space are plenty in number. Green Space symbolize peace, minimal stress and a cleaner environment for many people, which are considered as important factors in making a city livable, pleasant and attractive for its citizens. The primary functions performed by Urban Green Space are given in Table below.

S No.	Type	Functions
1	Regulation functions	Regulates the chemical composition of atmosphere and purifying the local air
		Controls the runoff and flooding
		Regulates the hydrological cycles
		Supports biological diversity in the city
		Prevents the soil erosion and sediments
2	Carrier functions	Regulates the local and global climate
		Conserves the energy in the city through control of the micro climatic variations
		Help recreation and tourism
3	Production function	Integrates urban man to nature.
		Recharging the ground water table
		Providing medical resources
4	Information function/ dissemination	Providing Raw material for some of the human activity
		Esthetic information
		Spiritual and religious information
		Cultural and artistic inspiration
		Scientific and educational information sources

Source :Role of Green Spaces for Maintaining Well-Being in Residential Community Development By Prashanti Rao <https://www.intechopen.com/online-first/76802>



BENEFITS OF GREEN SPACES: THE ECOSYSTEM SERVICES PERSPECTIVE :



Studies have been developed in relation with green spaces, that seek to understand the positive effects green spaces have on both the population around them and the city as a whole. The effects these areas provide are numerous, and relate to a wide range of different aspects of the city. The effects depend on the landscape qualities of the green space, such as the size, the shape, the placement and the content of the area. This approach is used to highlight the benefits of green spaces have in relation with the population, as well as categorize them in three main groupsThe benefits green spaces offer have been classified following this distinction of ecosystem services:

➤ PROVISIONING SERVICES:

The provisioning services, that include aspects such as food ,water etc., material benefits that the ecosystem.

➤ REGULATING SERVICES:

The regulating services is the services that regulate the ecosystem, and formulate the essential background for the development of other ecosystem services. .

- These green spaces, along with the vegetation that usually accompanies them, can help reduce noise levels in the city, and depending on the vegetation used.
- It can absorb or soften noise generated from different sources, such as traffic or the development of other activities within the city, such as industries.
- The presence of this vegetation can also improve air quality, as the plants will filter the air and reduce the amount of certain pollutants present in it.
- Vegetation can reduce heat islands by providing shadow and also reduce energy demand.

➤ CULTURAL SERVICES

This services are those immaterial benefits that citizens can obtain from the ecosystem. Cultural ecosystem services are considered to be the most important ones for city inhabitants. These areas are places that can also improve social cohesion, by providing a safe and close space for neighbors to meet, increasing therefore the sense of community, as well as the sense of safety, of well-being, of happiness, etc. green spaces must be maintained, monitored and offer varied facilities, improving the sense of security they produce, so they can reach their full potential and increase the benefits they provide, otherwise they might have a negative effect on people



Source: Evaluating the Urban Green Space benefits and functions at macro, meso and micro level: <https://www.ijert.org/research/evaluating-the-urban-green-space-benefits-and-functions-at-macro-meso-and-micro-level-case-of-bhopal-city-IJERTV3IS060563.pdf> Green spaces in cities <file:///C:/Users/90148/Desktop/clg/thesis/FULLTEXT01.pdf> Analysis and comparison of green spaces in Stockholm and Madrid Mónica Verduras-Stockholm university





ENVIRONMENTAL BENEFITS:

➤ ECOLOGICAL BENEFITS:

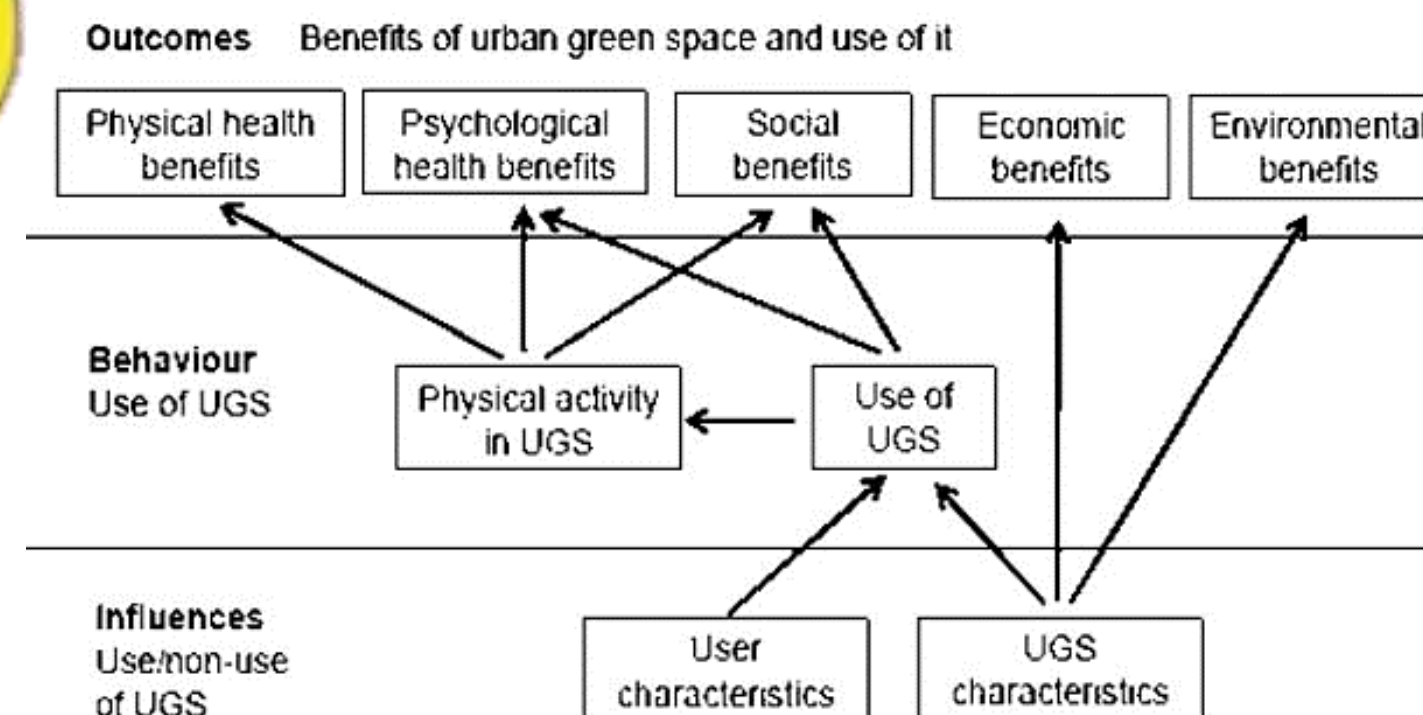
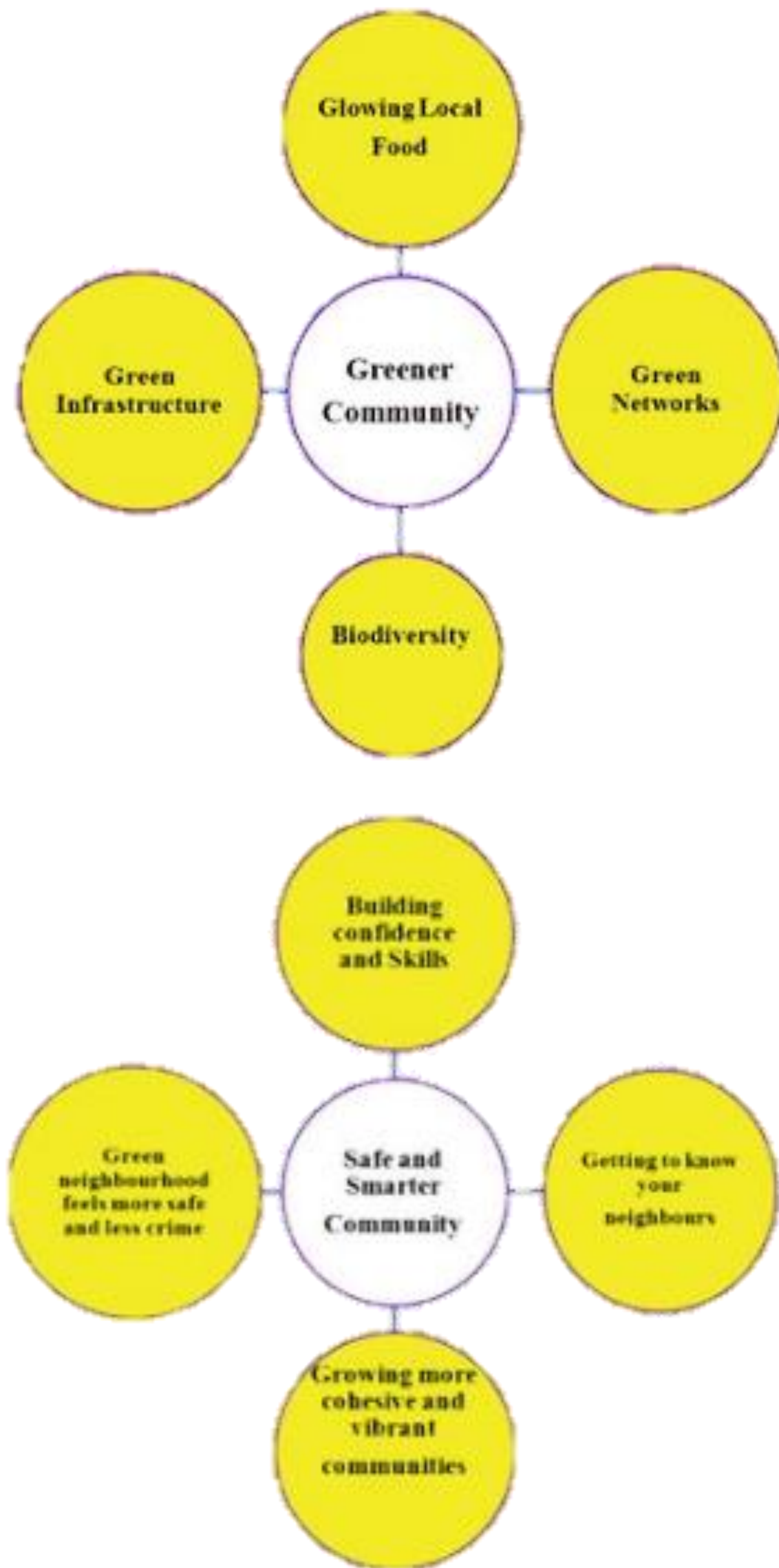
- urban green spaces supply cities with ecosystem services ranging from maintenance of biodiversity to the regulation of urban climate.
- Pollution control: green space may filter air, remove pollution and attenuate noise.
- Biodiversity and nature conservation: green spaces do functions as protection centre for reproduction of species and conservation of plants, soil and water quality
- They contribute to the healthy urban environment by providing clean air, water and soil.
- UGS provide safe play space for children & contribute to their physical, social and mental development.

➤ ECONOMIC BENEFITS:

- By increasing green space and tree planting. This provides a cooling effect and help to lower air temperature. And reduce energy cost for cooling buildings
- Property value: areas with enough greenery are aesthetically pleasing and attractive to residents, visitors and investors.
- Urban green spaces can be one of the factors that attract significant foreign investments that assist in rapid economic growth.

➤ SOCIAL AND PSYCHOLOGICAL BENEFITS:

- green spaces are key element for recreation and to experience nature.
- They enhance the quality of life for people living in the surrounding area and provide locations for a variety of leisure activities as a multi-use recreational open space.
- Green spaces are concerned with social values and Reduces crime, Strengthens community
- They enhance cultural life by providing venues for local festivals, civic celebrations and theatrical performances.



The recent model for the benefits of urban green space as shown in Figure suggested by Bedimo-rung et al. [8] describes the relationships between park benefits, park use, and physical activity, and the antecedents/correlates of park use. In this classification scheme, the discussion focuses on park environmental characteristics that could be related to physical activity, including park features, condition, access, esthetics, safety, and policies which is responsible for well being of residents.

Source: Green Space.doc <https://wichita.kumc.edu/Documents/wichita/wire/Green-Space.pdf> ARTICLES A Challenge to Cities: How Can We Incorporate Green Spaces? <https://www.arch2o.com/urban-green-spaces-challenge-cities/>



URBAN GREEN : INTRODUCTION BENEFITS CATEGORIES:

Urban green structures include a wide range of different components. Apart from parks, these include woodland, street tree and square plantings, cemeteries, private gardens, green roofs, community and allotment gardens, sports complexes, and so on. For the purpose of this review, we defined ‘urban park’ as Urban parks are defined as delineated open space areas, mostly dominated by vegetation and water, and generally reserved for public use. Urban parks are mostly larger, but can also have the shape of smaller ‘pocket parks’. Urban parks are usually locally defined (by authorities) as ‘parks’

Many scientific studies on urban green space start with stressing the multiple benefits of parks and other green areas There is general agreement, at least within the green space sector, that urban parks are essential for liveable and sustainable cities and towns.

The present report has one clear focus: documentation of the current scientific evidence for urban park benefits. Thus it will be possible to say, after reading each of the results section for individual benefits, if the best available, most current scientific evidence for the benefit is weak, moderate or strong. The report provides some insight in specific subthemes and individual studies

CATEGORIES OF URBAN PARK BENEFITS:

Urban park benefits, we are concerned with the services provided by the park that promote human or societal wellbeing, either directly or indirectly. According to Defra (2007), wellbeing is defined as a “positive, social and mental state; it is not just the absence of pain, discomfort and incapacity. It requires that basic needs are met, that individuals have a sense of purpose, that they feel able to achieve important personal goals and participate in society. It is enhanced by conditions that include supportive personal relationships, strong and inclusive communities, good health, financial and personal security, rewarding employment, and a healthy and attractive environment”.

The following potential benefits of urban parks were included:

- Human health and wellbeing, i.e. positive impacts of parks and park use on human health (both mental and physical) and wellbeing, either through direct or indirect effects such as recreation and leisure activities.
- Social cohesion / identity: the role of urban parks in strengthening social ties, relations and cohesion
- House prices: the value of urban parks as part of the living environment as reflected in higher real estate prices (for both houses and apartments).
- Biodiversity: the role of parks in harbouring and promoting biodiversity, and species diversity in particular
- Air quality and carbon sequestration: positive impacts of urban parks in terms of reducing air pollutant levels and carbon sequestration.
- Water management: contributions of parks to stormwater / run off regulation

ENVIRONMNETAL BENEFITS OF URBAN GREEN :

Urban greening offers improvements in air, water, and land resources by absorbing air pollutants, increasing water catchment and floodplain surfaces, and stabilizing soils. Urban forests act as temperature buffers providing shade in the summer, and wind break in the winter in addition to reducing noise pollution and CO 2 levels, and providing a habitat for wildlife. The economic benefits include marketable timber and agricultural products, as well as a variety of nontimber forest products such as artisan wares and honey.

The overall benefits to society, particularly to low-income residents, are significant. They include the contribution of trees and vegetation to the mental and physical health of the populace, and the provision of recreational opportunities and an outdoor classroom for environmental education. In addition, they provide aesthetic improvements to an environment otherwise dominated by asphalt and concrete

ENVIRONMNETAL BENEFITS :

- ❖ Air quality improvement

❖ Climate improvement

❖ Energy saving
- ❖ Solid Waste and Land Reclamation

❖ Noise Abatement

❖ Waste water treatment

❖ Erosion control

➤ **AIR QUALITY IMPROVEMENT:**

While air pollution indices in many cities in more developed countries have dropped over the last ten to twenty -five years, air pollution levels have been rising in Cities. The average level of particulate suspension in the atmosphere rose from 65 mg/m 3 to 400mg/m 3 be tween 1974 and 1990 (Carter 1993).

- Urban greening can reduce air pollutants to varying degrees. Air pollution is directly reduced when dust and smoke particles are trapped by the vegetation.
- Urban vegetation can reduce carbon dioxide levels in two ways. First, all plants, through photosynthesis, ab sorb carbon dioxide directly into their biomass and release oxygen in return.
- extensive vegetation cover reduces the A heat island effect in an urban area, residents use fewer fossil fuels to cool buildings, there by reducing power plant emission.

➤ **CLIMATE IMPROVEMENT:**

One of the most important benefits of urban vegetation is its impact on the climate. Two distinct influences can be identified.

- The direct effect on human comfort. Secondly, there is an effect on the energy budget of buildings in cities where air-conditioning is used.
- Trees can significantly increase human comfort by influencing the degree of solar radiation, air movement, humidity and air temperature and providing protection from heavy rains.
- urban heat island effect where a city heats up rapidly and then maintains a high temperature.

➤ **ENERGY SAVINGS:**

Various components of an urban greening program can contribute substantially to reducing a cities overall energy budget. As mentioned previously, inner city concentrations of pavement and concrete produce an urban heat island effect that causes discomfort and health problems

- By increasing green space and tree planting. This provides a cooling effect and help to lower air temperature. And reduce energy cost for cooling buildings
- Increasing tree cover in the city by 10 percent may reduce the total energy used for heating and cooling by 5 to10 percent.
- Urban green areas can supply fuelwood and other substitutes for imported fossil fuels, thereby providing sources of renewable energy.

➤ **NOISE ABATEMENT:**

Noise often reaches consistently unhealthy levels in large cities like Hyderabad. To make matters worse, the building materials used in low-income neighborhoods do not insulate the residents from noise as well as the more substantial materials frequently used in wealthier neighborhoods.

- leaves, twigs and branches will absorb sound, as will grasses and other herbaceous plants. Tree or plant barriers will deflect sound away from listeners and, if at right angles to the source, will reflect it back to that source
- Vegetation can also mask sounds to the extent that people will filter out unwanted noise by selectively listening to the sounds of nature (bird songs, leaves rustling, etc.) over the sounds of the city.

➤ **EROSION CONTROL:**

The lack of vegetative cover and the hard seasonal rains most of these cities experience, erosion and landslides have become common occurrences. Informal settlements are hit especially hard by landslides, as they are most often built on marginal slopes. Much of that risk can be reduced by planting hardy species to hold eroding soil on slopes.

➤ **WASTEWATER TREATMENT:**

The practice of at least partially treating waste water in stabilization ponds has to be considered an alternative for urban waste water treatment as much as its economic and ecological advantages. Having significant wastewater treatment could help stabilization of ponds, integrated into park systems, would economically and ecologically serve some of their treatment needs. Using the parks retention ponds and artificial wetlands to partially treat its waste water. Instead of being whisked away through concrete channels (as occurs in many cities), the water entering this park area slowly passes through a series of natural watercourses with floodplains and wetlands.

➤ **SOLID WASTE AND LAND RECLAMATION:**

Solid waste disposal in most large cities has become a serious and permanent problem. While some cities have managed to keep pace with their growing populations, the majority suffer from significant garbage dilemmas, especially in and around informal settlements. Urban greening can help abate the solid waste disposal dilemma. Many forms of waste and nutrient recycling exist and are already in use in other parts of the world. settling pond sludge is used for agricultural fertilizer;and wastewaters are recycled into irrigation for aquaculture and agriculture.

Source : Good Practices for Urban Greening <https://publications.iadb.org/publications/english/document/Good-Practices-for-Urban-Greening.pdf>



SOCIAL BENEFITS:

Green spaces help in improving the health of the public, as it is discovered that time spent in nature, relieves mental fatigue and the feelings of violence and aggression that can spring from it. Natural resources provide many “activities” that require little to no effort, yet provide ways to restore a person’s health and mental wellbeing. This kind of green spaces can help to relax

- Health
- Employment
- Recreation
- Education
- Aesthetics

➤ **HEALTH:**

The benefits of urban greening to humans health can be considerable improvements in air quality due to vegetation have positive impacts on physical health with such obvious benefits as decreases in respiratory illnesses.

- The fact that urban forests also reduce stress and improve health by contributing to an aesthetically pleasing and relaxing environment.
- Urban forests provide a connection between people and their natural environment. This connection is important for everyday enjoyment, worker productivity and general mental health.
- Shade from trees and the resulting cooler temperatures, Shade also reduces ultraviolet light exposure thereby lowering the risks of harmful health effects such as skin cancer and cataracts

➤ **EMPLOYMENT:**

Another important aspect of urban greening is the jobs provided for poor, skilled and unskilled laborers. Urban greening projects are often labor-intensive and provide both initial start-up jobs (soil preparation, planting, etc.) as well as more permanent employment (maintenance, management, etc.). Project managers have estimated that the program will need approximately 3,380 workers to produce and transport plants, 3,700 working in the plantations, 800 in management and more than 100 for protection and surveillance in existing green areas.

➤ **RECREATION:**

Green areas provide recreational sites, especially for lower income residents who tend to frequent city parks more than wealthier citizens because of financial constraints and restrictions on leisure time. The urban poor generally have few affordable options for recreation, and thus place a high value on green areas. Nevertheless, their preferred recreational activities may vary from city to city, or even among neighborhoods

➤ **EDUCATION:**

- Parks and other green areas also provide educational opportunities for urban residents. Individuals, families and school groups can take advantage of a city's green areas to learn about the environment and natural processes. For urban children, as well as adult students, the learning experiences available in urban parks may be some of the few opportunities they have to learn about nature through first-hand experience.
- Moreover, by getting the public involved in educational activities associated with urban green spaces, planners can raise the consciousness of the public concerning the importance of these spaces.
- Another way to educate the public on the importance and benefits of urban greening is to involve people in the greening process itself.

➤ **AESTHETICS:**

- While not considered as important as filling basic needs such as food and shelter, the aesthetics of green areas can also be very meaningful to many urban residents. Vegetation reduces sun glare and reflection, complements architectural features and tones down the harshness of large expanses of concrete.
- Areas of a cities with enough greenery to be aesthetically pleasing, are attractive to residents and investors alike. The beautification of Singapore and Kuala Lumpur, Malaysia, was one of the factors that attracted significant foreign investment that assisted those cities rapid economic growth.
- Another advantage of aesthetically pleasing green areas is their positive effect on property values. When unsightly vacant lots or garbage dumps are replaced by attractive parks, not only does the residents= quality of life improve, but the value of their property increases.
- In addition, rehabilitating lands with vegetation is often more attractive and cost-effective than constructing new buildings on them. Urban greenery, used as a boundary line, can improve the aesthetics of small private landholdings in urban or suburban areas.

Source : <https://www.iadb.org/> Good Practices for Urban Greening Washington, D.C. May 1997 C ENV-109 - <https://publications.iadb.org/publications/english/document/Good-Practices-for-Urban-Greening.pdf>



URBAN GREEN GROWTH STRATEGIES FOR INDIAN CITIES

India has to leverage its cities despite the fact that an overwhelming majority of them are deficient in basic services, financially weak and lacking the capacity required to plan and implement change. Growing at a decadal rate of 31.76% (2001-2011), these urban areas house almost 377 million people and are expected to accommodate 590 million people by 2030. Urban growth is far exceeding the capacity of infrastructure and services, and inadequate environmental management measures have contributed to a significant degradation of valuable natural resources, adversely affecting the quality of life of urban dwellers. Conventional urban planning and management tools have become barriers in their development and in order for this growth to be sustainable, there is a desperate need for new policies and implementation strategies that can generate positive social, environmental and economic impact.

- About 27.8% of population are residing in urban areas in India.
- There are 6 mega cities (4 million plus) & 35 metropolitan cities (1 million plus). There are also, 393 large towns (0.1 million plus) and 3,784 small and medium towns (less than 0.1 million population).
- In the context of carbon offsets, massive greening is required in urban areas to provide shade and insulation to buildings. Urban greening plays a vital role in the amelioration of urban environment. population increment of 90.5 million for the first time since independence (Bhagat, 2011) and is expected to hit figure of 550 million or 42% of the total population by 2030 (Roberts and Kanaley, 2006).

Table 1: Table Showing Pattern of Urbanization in India

Census Year	Urban Population (in millions)	Percentage Urban	Annual Exponential urban growth rate
1961	78.94	17.97	-
1971	109.11	19.91	3.23
1981	159.46	23.34	3.79
1991	217.18	25.72	3.09
2001	286.12	27.86	2.75
2011	377.10	31.16	2.76

Source: Bhagat, 2011

With growing cities the demand of resources like water, land etc. has grown proportionally to the growing rate of the urban population. Hence, Cities are experiencing increasing signs of environmental stress, notably in the form of poor air quality, excessive noise, and traffic congestion (Ruangrit & Sokhi, 2004). At the same time, irregular and unsustainable extension of cities has caused destruction of urban green areas and resulted from increasing demand for land

MAJOR INDIAN CITIES WITH PER CAPITA GREEN SPACE.

The major impact of urbanization on landscape surfaces is the replacement of vegetated areas with artificial surfaces that are mostly impervious. These impacts modify the surface energy balance through a change in absorption and reflection of solar radiation. Since 1970, India has been undergoing rapid urbanization. The urban population rose from 109 million in 1971 to 377 million in 2011, by an 11.7% increase (from 19.9% to 31.6%) over four decades. The number of cities in the country with the population exceeding one million has steadily increased from 23 in 1991, and 35 in 2001, to 53 in 2011.

For instance, New Delhi, a megacity in a humid sub-tropical setting, has complex patterns of urbanization in terms of its commercial, residential, mixed-use areas, and traffic intersections. For this, the major contributing factors are high population density, high density of road network, and a very high amount of traffic flow. Unlike Delhi, the city of Pune is located in a hot, semi-arid region with village nuclei and industrial sectors. Pune is affected by a sudden recent increase in population by 10 times compared to the last century

❖ Table shows the population of populous cities of 2018 Census which was collected from the India Environment and Population Portal [34]. The table also provides information about the geographical area, forest and tree cover, and per capita green space in these cities

Table 1. Major cities of India with per capita green space.

City	Population in Millions	Population Density (km ⁻²)	Geographical Area (km ²)	Green Cover % (in km ² ; 2017)	Per Capita Green Space (m ² ; 2018)
Delhi	28.50	12,591	1484.00	20.00 (296.80)	10.41
Mumbai	23.50	20,482	603.00	36.48 (220.00)	9.36
Kolkata	15.20	24,400	1380.00	7.30 (100.74)	6.61
Bangalore	13.90	4381	2196.00	2.09 (46.03)	3.31
Hyderabad	11.57	18,480	650.00	1.66 (10.79)	0.93
Chennai	9.88	14,350	1189.00	15.00 (178.35)	18.05
Ahmedabad	8.41	9900	464.00	17.00 (78.88)	9.38
Surat	6.55	1376	326.50	11.84 (38.66)	5.90
Gandhinagar	6.33	660	649.00	54.00(188.46)	29.77
Jaipur	3.71	598	467.00	5.43 (24.75)	6.67
Nagpur	2.94	11,000	285.90	18.00 (51.42)	17.49
Mysore	1.70	6911	128.40	20.19 (25.92)	15.25
Chandigarh	1.05	9252	114.00	35.00(39.90)	38.00

Note: Figures in bold indicate the above optimal of 9 m² per capita green cover.

- ❖ It can be noted from that many cities in India already fall short of green space available per capita, which is much below the WHO recommended norms of 9 sq m/capita.
- ❖ Global standards recommend 33% green cover for urban areas, and that the maximum green spaces be connected.
- ❖ The LEED ND (Leadership in Energy and Environment Design Neighborhood development) recommends that the green area per capita be >20 sq m or maintain a minimum of 1.25 ha of open space per 1000 residents, and that the access to open space be within 250 m of residential areas for smaller parks

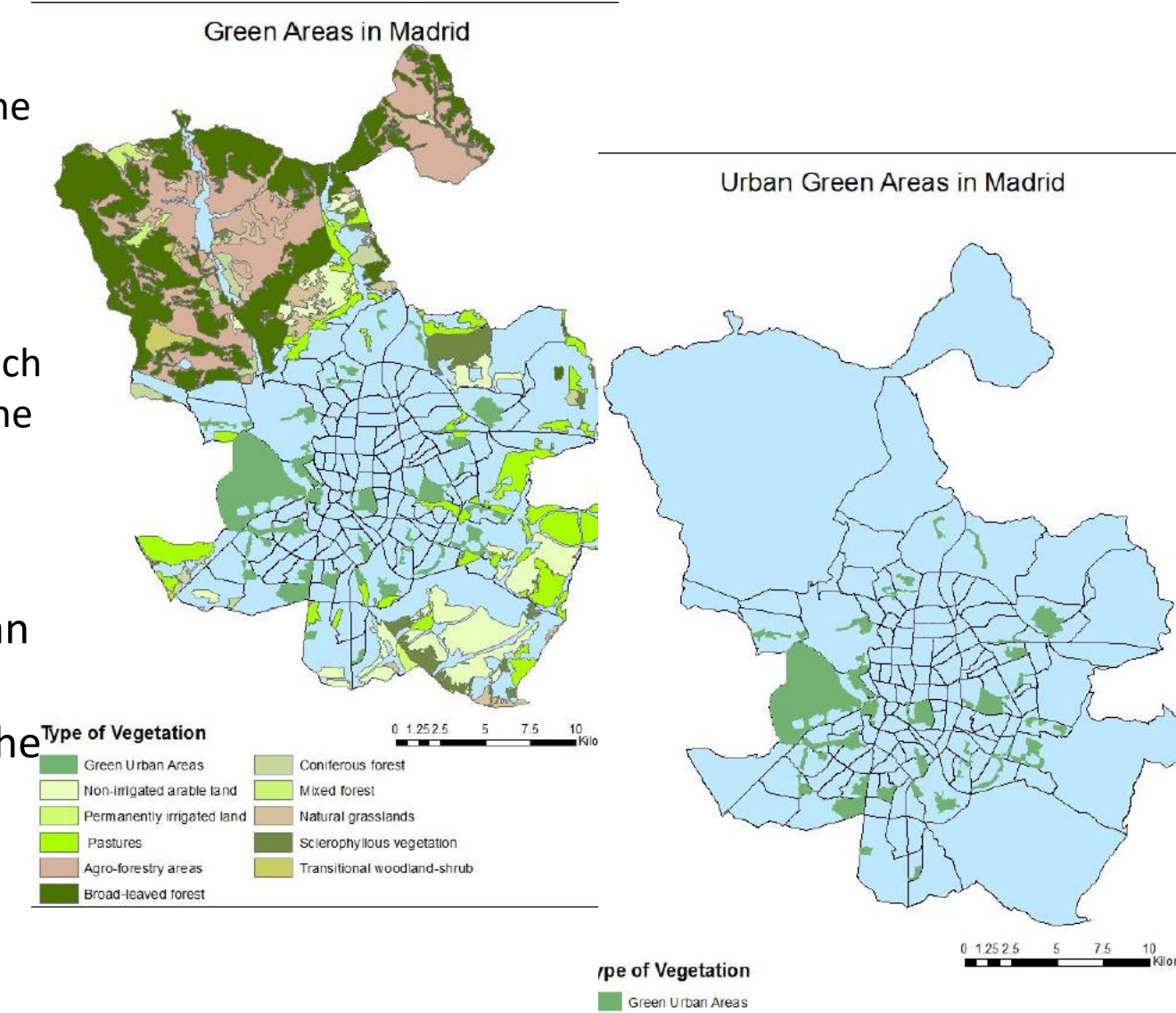
History of Green Spaces in Madrid:

Madrid is a city located in the mountainous region of the center of Spain. The city is characterized by the natural areas around its limits, parts of bigger green structures. This green natural areas are part of protected natural parks: On the northwest of Madrid, the area of El Pardo and Soto Viñuales; and on the southeast, the Parque Regional del. The presence of these green natural areas of such a large ecological value within the limits of the city is a uniqueness of Madrid hard to find in another cities of Europe. The rest of green spaces within the city are situated in the urban area, located in outer areas of the city, leaving the inner city 'empty' from these spaces. Madrid presents an interesting mixture of green spaces created recently, following modern trends and designed with a clear social purpose, and parks and green areas that were created only for royal recreation. The biggest green urban areas in the city belong to this last group of spaces. These areas, such as the Retiro park, or Los Jardines del Capricho, are catalogued as 'historic parks' in the legislation and were opened to the general public at the ending of the 19 century. These areas have the particularity that have kept their historical values from their design, since they were protected and couldn't be modified while they were still under the control of the Crown.

Green Spaces in planning practices in Madrid:

The planning system in the city of Madrid is defined by the General Plan of Urban planning of Madrid. This plan was developed in 1997, and has been applied since then, with the addition of some modifications to the original regulations. Nevertheless, the main ideas present in this plan have been mostly unchanged until nowadays. A new document in regards to the planning of green spaces was designed and completed in 2013, the Plan Revision for Urban Sustainability. Including of the green regions , developing the green spaces the quality of the urban landscape

In addition, it is important to mention that green spaces within the city (also sometimes called 'parks') are classified in different groups according to their values, year of development, etc. These categories are: Urban parks (Parques rbanos), Historical Parks (Parques históricos) and Forestal parks (Parques forestales), each with specific conservation and planning regulations.



Green spaces in Stockholm

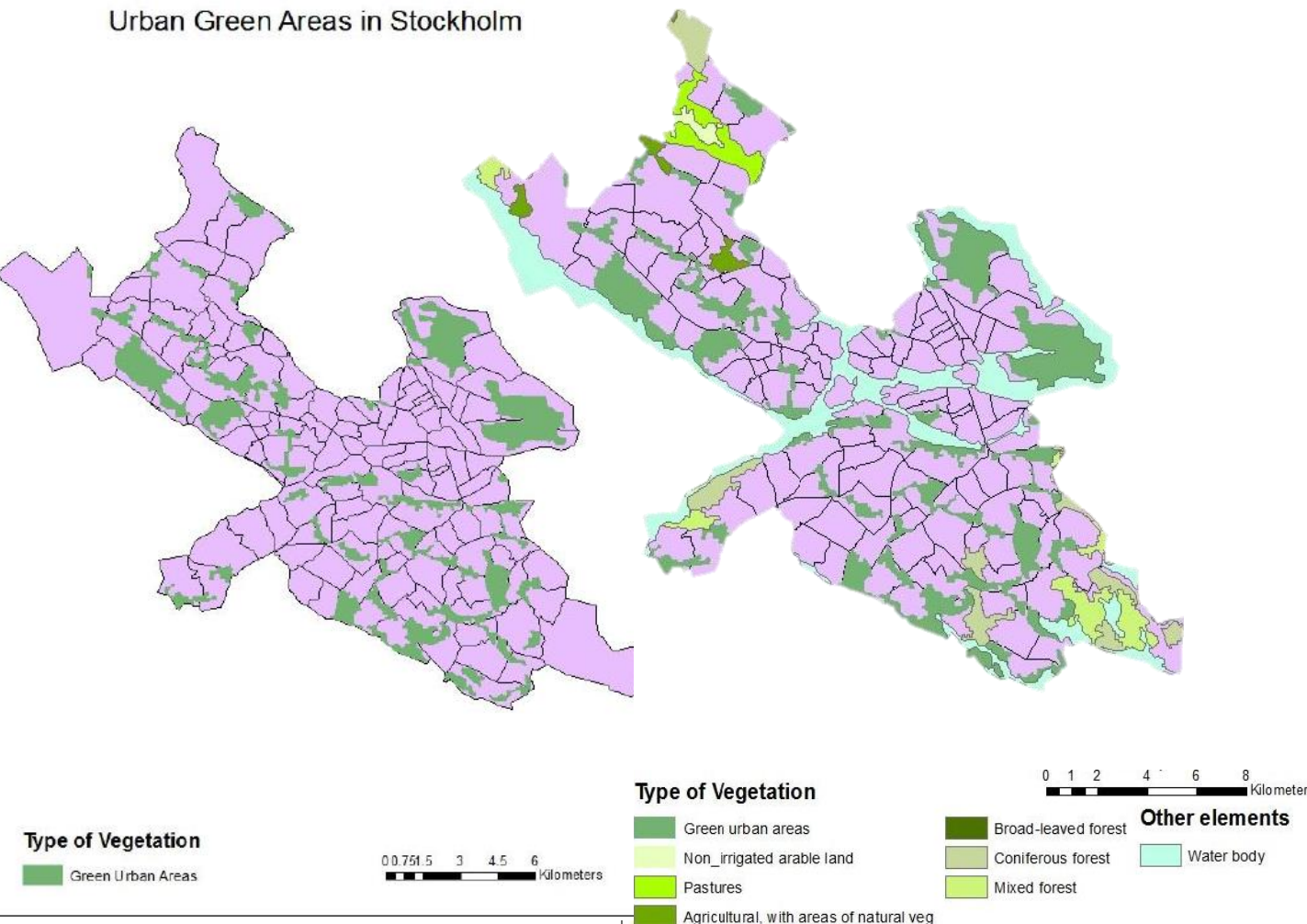
Background information about Stockholm:

The city of Stockholm has several green spaces that were created following this trends of growth, along with planning directives and currents of thought. Various of these examples can still be found in the city. Around 1700 the main inspiration for green spaces were the English landscape parks, characterized by natural- romantic forms, an example of this kind of park is Bellevue, located in Normalm district. From 1850 to 1880 public city parks were developed, with paths, trees and exotic plants. Around 1960-1970, late modernism was translated into the importance of large scale, with big grasslands and single minded plant materials. From 1980-1990 Stockholm was densified, and parks were established. These parks were larger green multifunctional spaces, that vary from natural parks, to intensively planned ones (Markkontoret, 2006). In addition, 'Kvartersparker' (or 'quarter parks') were also located inside the city. These parks are 'neighbourhood' green spaces, usually included within the housing areas (Markkontoret, 2006). These areas complement the presence of larger green spaces in the city, and have an important role as they are more directly connected with the population.

Green Spaces in planning practices in Stockholm:

Stockholm's City policies have been working towards a sustainable development, branding the city as 'green' within the international context. This aspect also affects that way in which green spaces are treated in this city. Following the guidelines set in Rio Earth Summit (1992), Swedish legislations were remodeled towards a more sustainable development, including the management of green spaces in and surrounding urban areas. The Swedish towns and cities are encouraged to develop green plans as an important part of the compulsory structure plan. The focus of this green plan is to identify urban green spaces and highlight their value to the community, in order to try to achieve sustainability and to preserve biodiversity. The plan provides the need to value the benefits green spaces provided as well as the need to guarantee for the future development has a positive impact. This plan established given amount of green spaces and the closeness to nature the city has access to the parks Sports facilities and much more green in side the city

Urban Green Areas in Stockholm

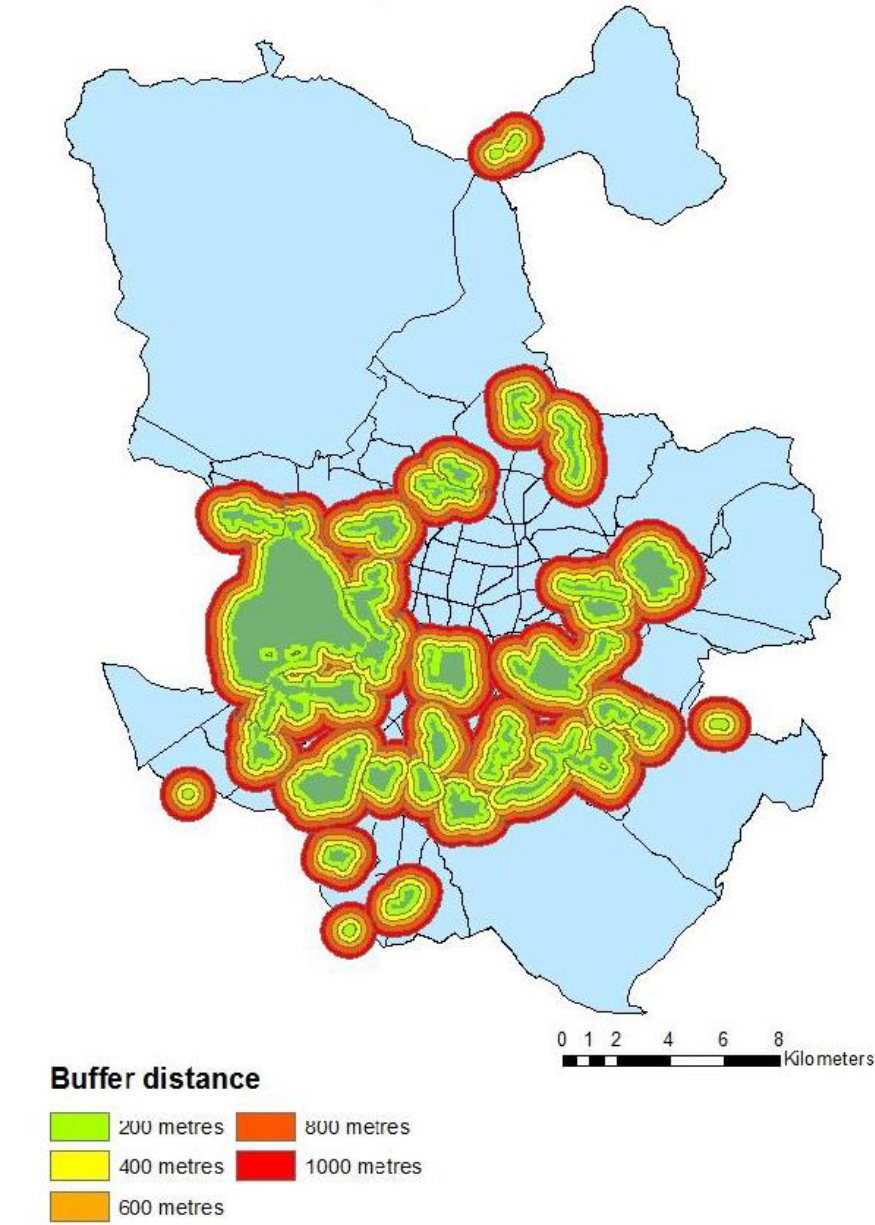


Result obtained from Madrid

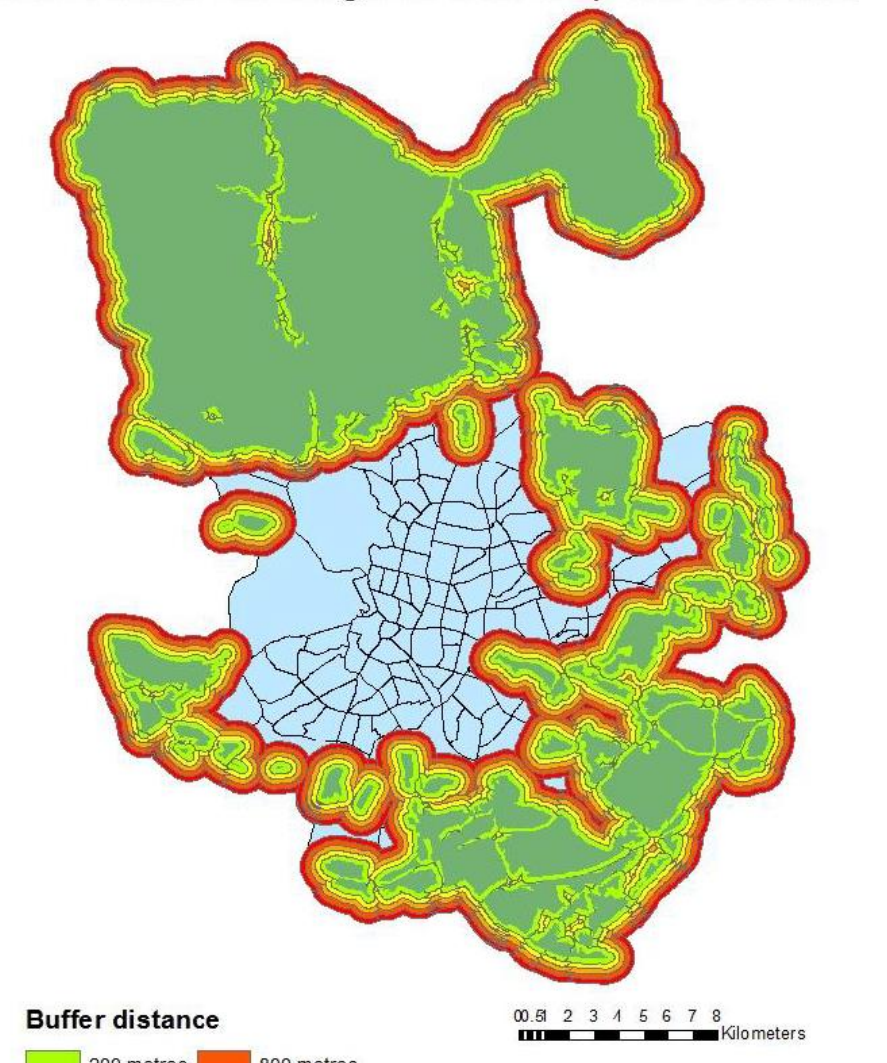
	Total Area (ha)	% Occupied by Green Spaces
Green Urban Areas (ha)	3573.389195	5.911767141
Green Natural Areas (ha)	28842.24064	47.71621594
Total Area Madrid (ha)	60445.36447	53.62798308

Fig 2. Figure showing the distribution of green spaces (both 'urban' and 'natural') in relation with the whole extension of the area of Madrid, in order to measure the 'quantity' of these areas in the city.

Buffer areas around green urban spaces in Madrid



Buffer areas around green natural spaces in Madrid

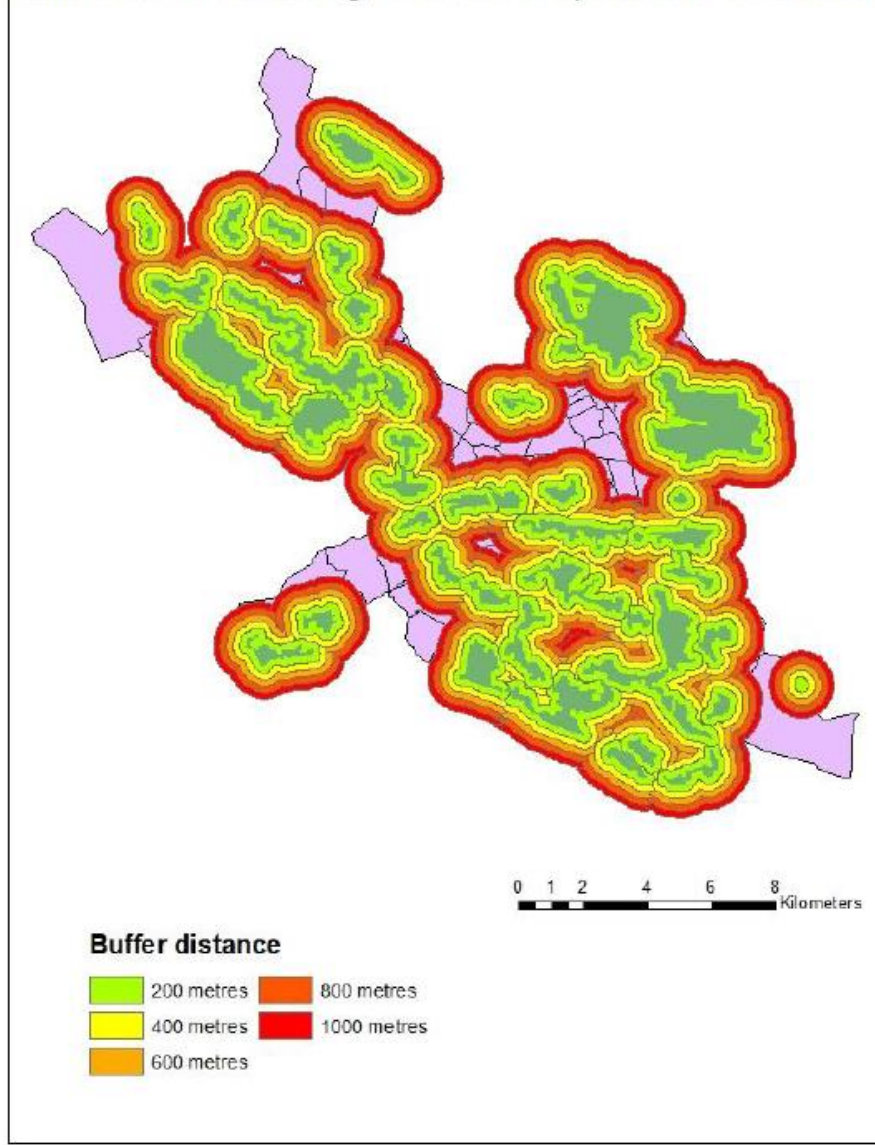


Results obtained from Stockholm

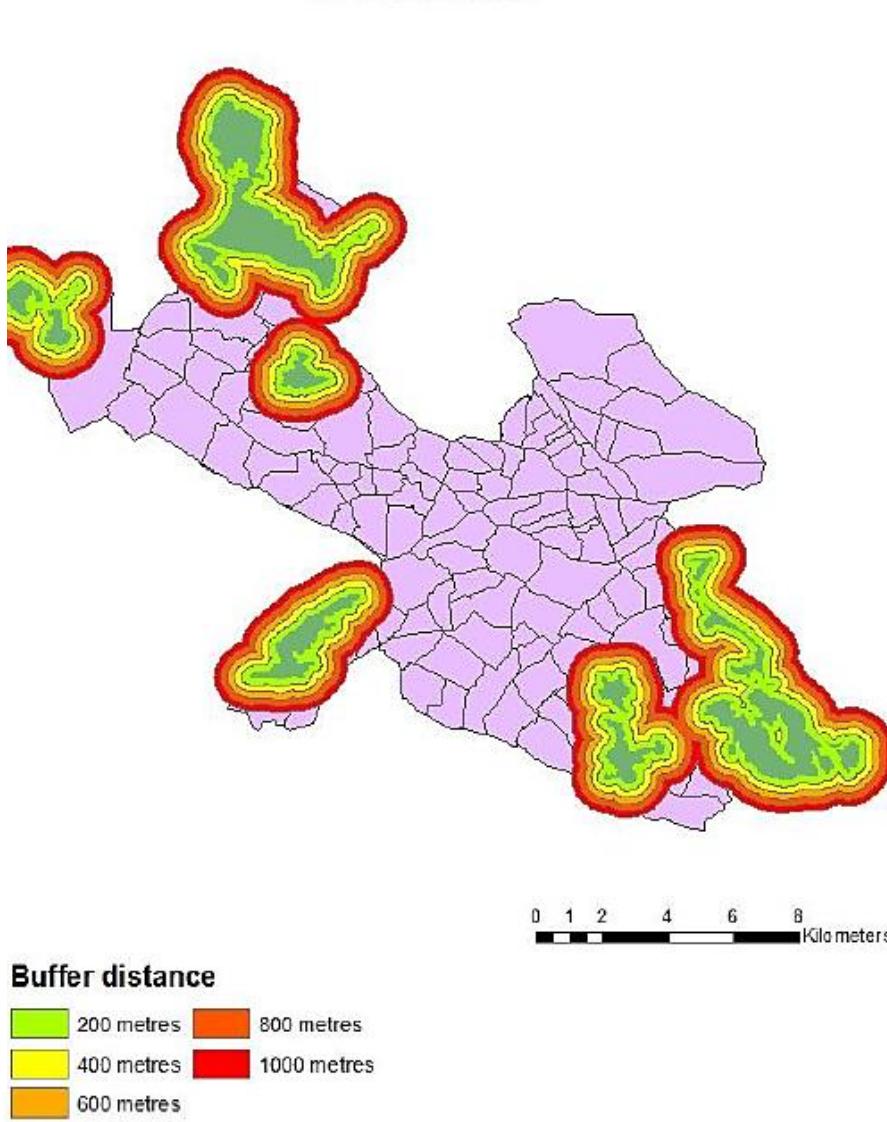
	Total Area (ha)	% Occupied by Green Spaces
Green Urban Areas (ha)	3723.457448	17.25333378
Green Natural Areas (ha)	2013.74	9.33103946
Total Area Stockholm (ha)	21581.08975	26.58437324

Fig 5. Figure showing the distribution of green spaces (both 'urban' and 'natural') in relation with the whole extension of the area of Stockholm (this is, their 'quantity'). The area occupied by water bodies was not included.

Buffer areas around green urban spaces in Stockholm



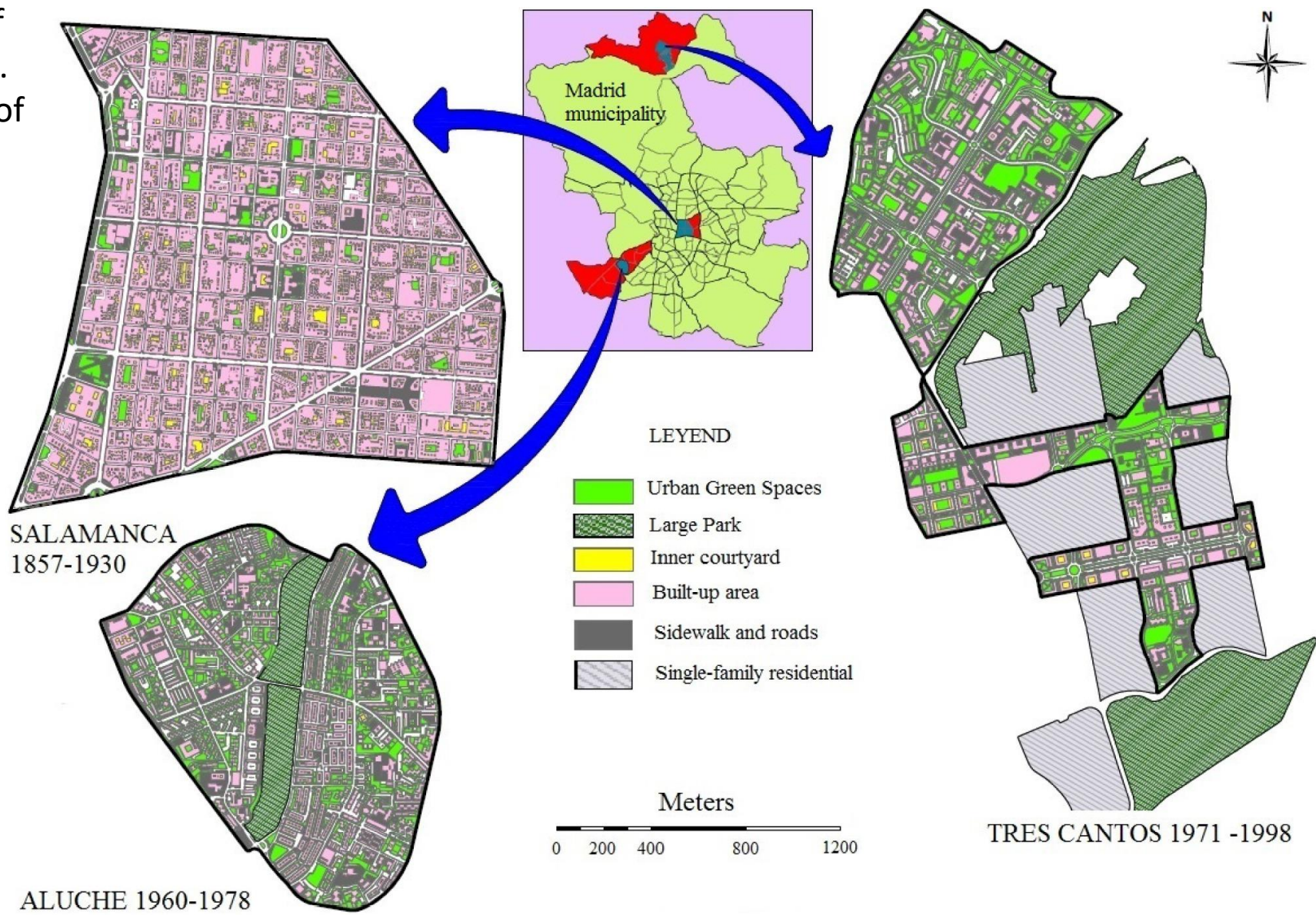
Buffer areas around green natural spaces in Stockholm



Map 15. Map showing the influence of green urban areas in the form of buffer areas in Stockholm, this is, their accessibility.

Map 16. Map showing the influence, in the form of buffer areas, of natural green spaces in Stockholm

Madrid has 300,000 trees and 6400 hectares of green space, which means it is the second most sustainable city after Tokyo. Many other countries are starting to turn towards more sustainable initiatives. They have found Madrid’s commitment to green living to be admirable and are taking their own steps. The access to the beautiful forests is one of the biggest reasons that Madrid has become such a green city. It has started to embrace other sustainable ideas as well. Monte de el Pardo, a protected natural park, is especially beautiful and filled with animals. It is about 50 times the size of Central Park in New York,



Good practices in madrid and stockholm Spain:

The Barcelona Green Infrastructure and Biodiversity Plan 2020 Barcelona is one of the most densely populated cities in Europe. Whilst it has a relatively low ratio of green space per inhabitant, it has more street trees than most other European cities: currently around 161,423 trees of 150 different species line the streets of Barcelona. The ‘Barcelona Green Infrastructure and Biodiversity Plan 2020 which was released in 2013, lists over 70 projects and actions

aims: provide environmental and social services, introduce nature into the city, increase biodiversity, increase connectivity among patchy Green Infrastructure and make the city more resilient

Benefits of the project include:

- Trees have a positive impact on health since air pollution is a major environmental challenge for the city;
- Health and educational benefits from gardening and contact with nature

Ongoing Green Infrastructure projects and biodiversity projects are: Green Corridors Network, urban allotments with eco-gardening, habitat conservation (artificial reefs) and species conservation (protection of amphibians in urban ponds and protection of urban birds). There are artificial reefs in the Mediterranean Sea south of Barcelona to restore marine biodiversity; this pioneer project aims at facilitating the reproduction of species and benefitting traditional fishing methods.

STOCKHOLM a sustainable green city:

Stockholm in brief:

Stockholm is known for its beauty and for its proximity to water. The City of Stockholm has been working determinedly on urban environment issues for a long time. In 2010 the European Commission designated Stockholm the first European Green Capital.

A growing city:

Stockholm is a fast-growing city. Space is needed for some 200,000 new Stockholmers by 2030. Planning is therefore already under way for around 30 major urban development projects and 100,000 new homes. This entails many challenges if the goal of being a long-term sustainable city is to be met.

A city for walking:

The general development plan for Stockholm, entitled ‘The Walkable City’ describes how the city will grow and develop, connecting the inner and outer districts. Entire environmental districts are also emerging in Stockholm, such as Hammarby Sjöstad and Stockholm Royal Seaport, and they are inspiring city developers from around the world.

Mobility strategy:

Stockholm invested in a metro system early on, and was one of the pioneers in introducing a congestion charge for road traffic in rush hours. Further measures are now needed to increase mobility and reduce impact on the climate, environment and residents’ health. One of these is a special focus on the bicycle as a means of transport.

The clean city:

Stockholm is primarily heated by district heating, which has dramatically reduced carbon dioxide emissions. The combined power and heating plants are fired with climateneutral fuels such as biofuel and household waste. Stockholm’s goal is to be fossil-fuel free by 2050



Stockholm – facts & figures

Population:
880,000
Area:
209 km², of which 21 km² water
City built on 14 islands
40% is parks and green areas



To develop in a sustainable way it is important to nurture the city’s many green areas and their values even though the city grows.



A CASE STUDY OF PUNE, INDIA

AIM : The main objective of the study to understand the social, ecological and economic values of urban green spaces. and deals with changes with urban Land Use / Land Cover Changes using remote sensing technique. For this purpose Landsat ETM+ images of 1999 and 2009 was obtained for the study area.

STUDY AREA

Pune City (Pune municipal corporation limit), lies between latitudes 18°25' N and 18°37' N and 73°44' E and 73° 57' E longitudes covering an area of 243.84 sq km. Situated on the Deccan plateau and

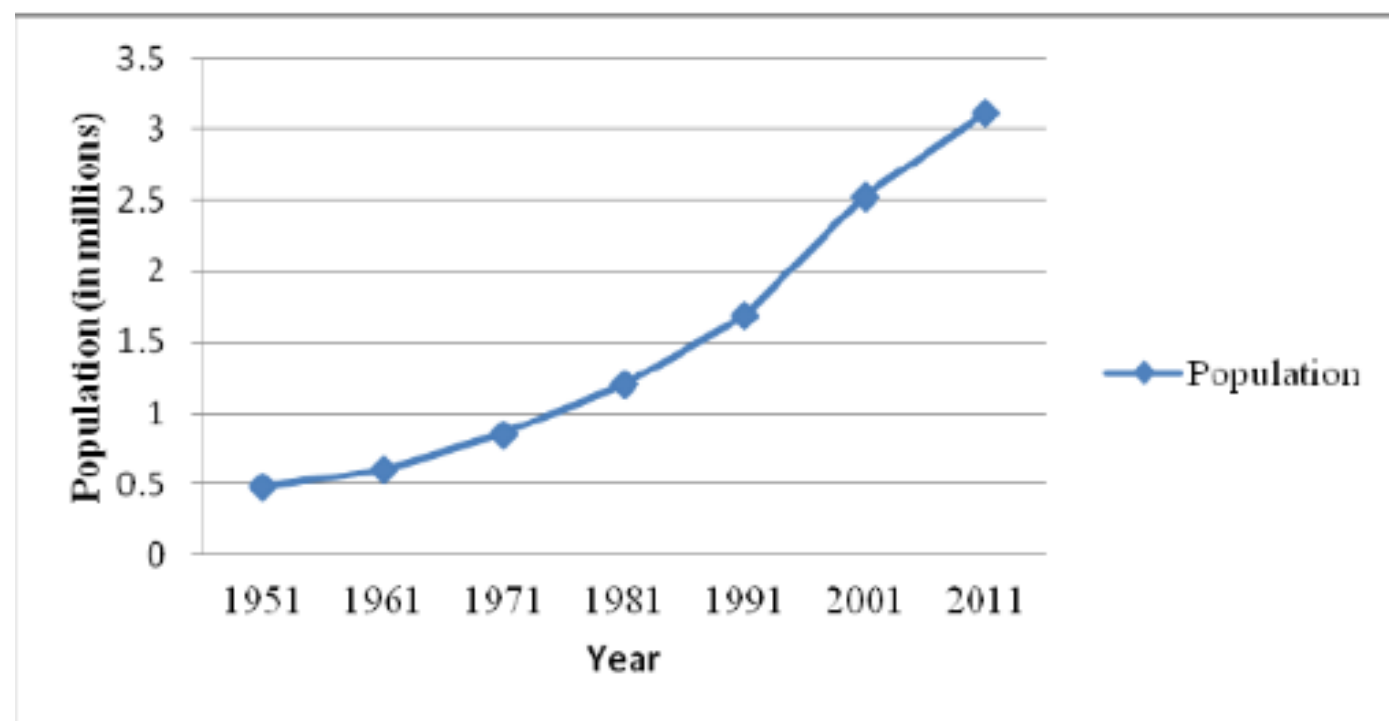


Figure 1: Demographic Trend of Pune City (Source: Census of India, 1951-2011)

Landuse Class	Area in 1999 (%)	Area in 2009 (%)	Area in 1999 (Sq. Km.)	Area in 2009 (Sq. Km.)	Area Difference (Sq. Km.)	Area Difference (%)
Water Bodies	1.47	1.20	3.58	2.93	-0.66	-0.27
Built Area	30.86	48.50	75.25	118.26	43.01	17.64
Barren & Fallow Land	36.20	21.80	88.27	53.16	-35.11	-14.40
Dense Vegetation	13.50	12.82	32.92	31.26	-1.66	-0.68
Sparse vegetation	17.97	15.68	43.82	38.23	-5.58	-2.29
Total Vegetation	31.47	28.50	76.74	69.49	-7.24	-2.97

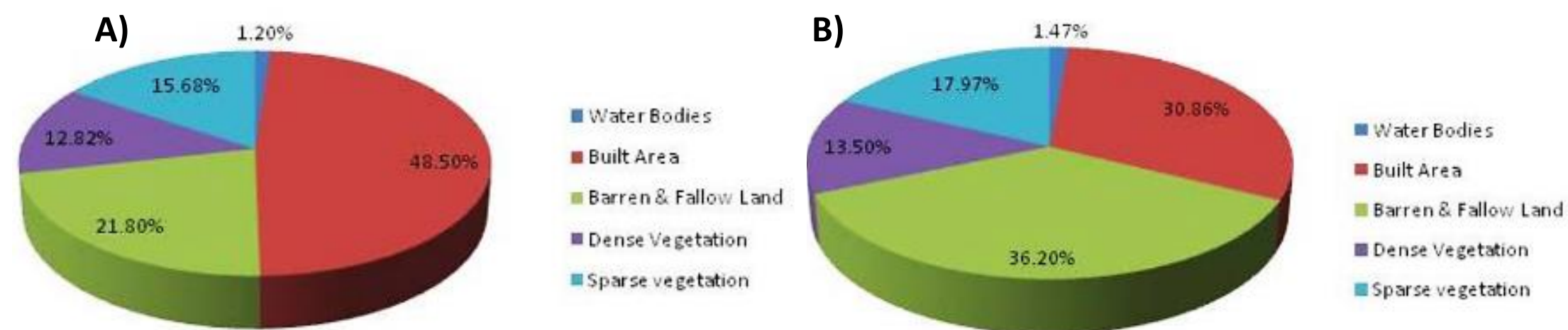
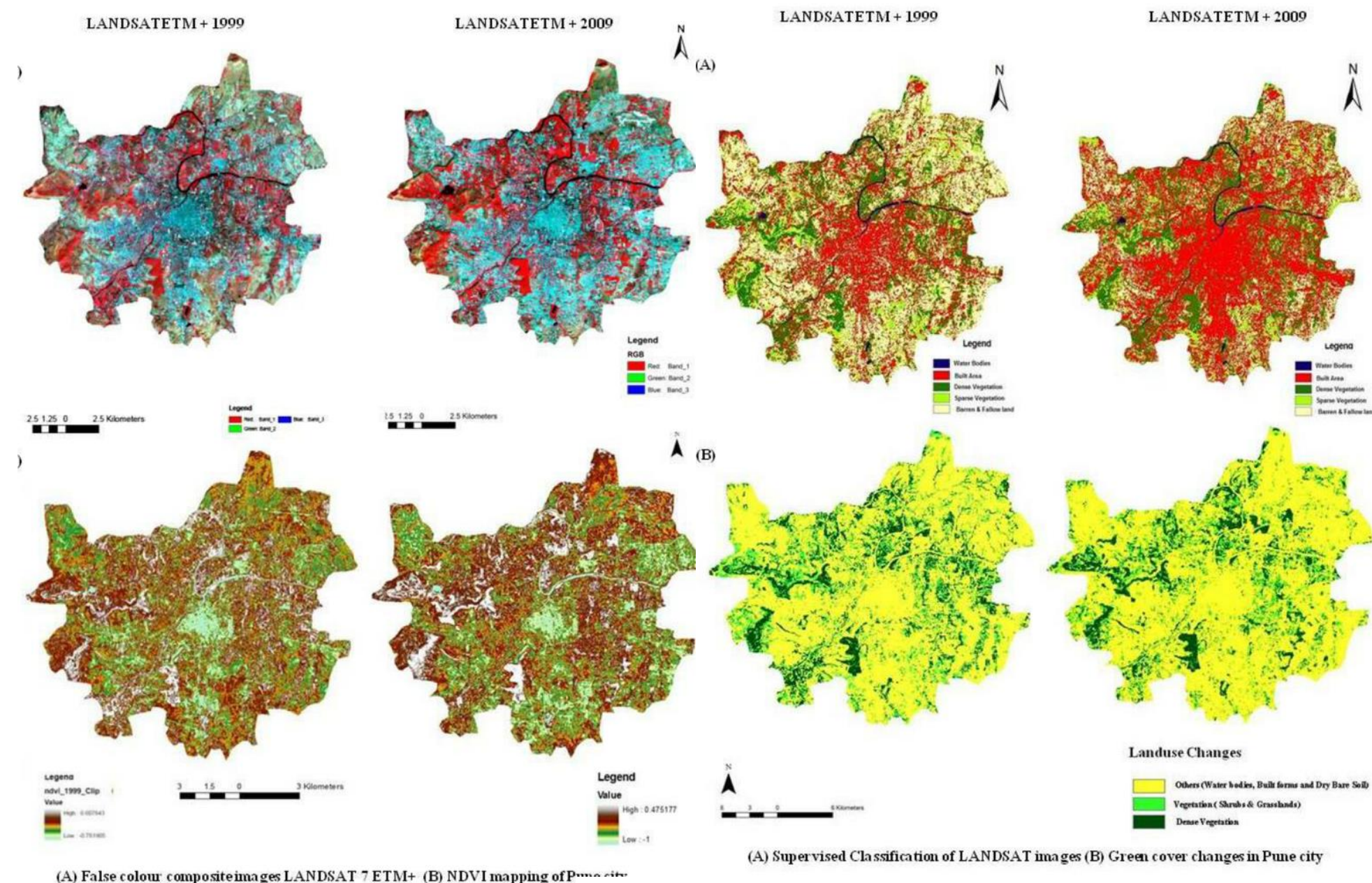
Pune's built up area has increased from 30.86% in 1999 to 48.50% in 2009 i.e. substantial increase of 43.01 sq km. of area in just last 10 years, on the other hand barren & fallow land area has decreased considerably from 36.20% in 1999 to 21.80 % in 2009.

Major objective of the study was to analyse the changing trend of the urban green spaces in

the city, it was noticed that both sparse and dense vegetation showed decrease in areas of 5.58 and 1.66 sq km. respectively in past 10 years. Thus, this analysis highlights that the city is under high pressure of urbanization and sprawl,

taking place by grabbing and converting existing habitat like agricultural, barren, fallow land &

forestlands into built habitat and grey infrastructure. The overall mapping accuracy of the land use/ land cover map was 86.73%



A) Land use/ landcover of pune 1999 B) land use and landcover of pune 2009
Graphical Representation Land use/Landcover Changes in Pune City

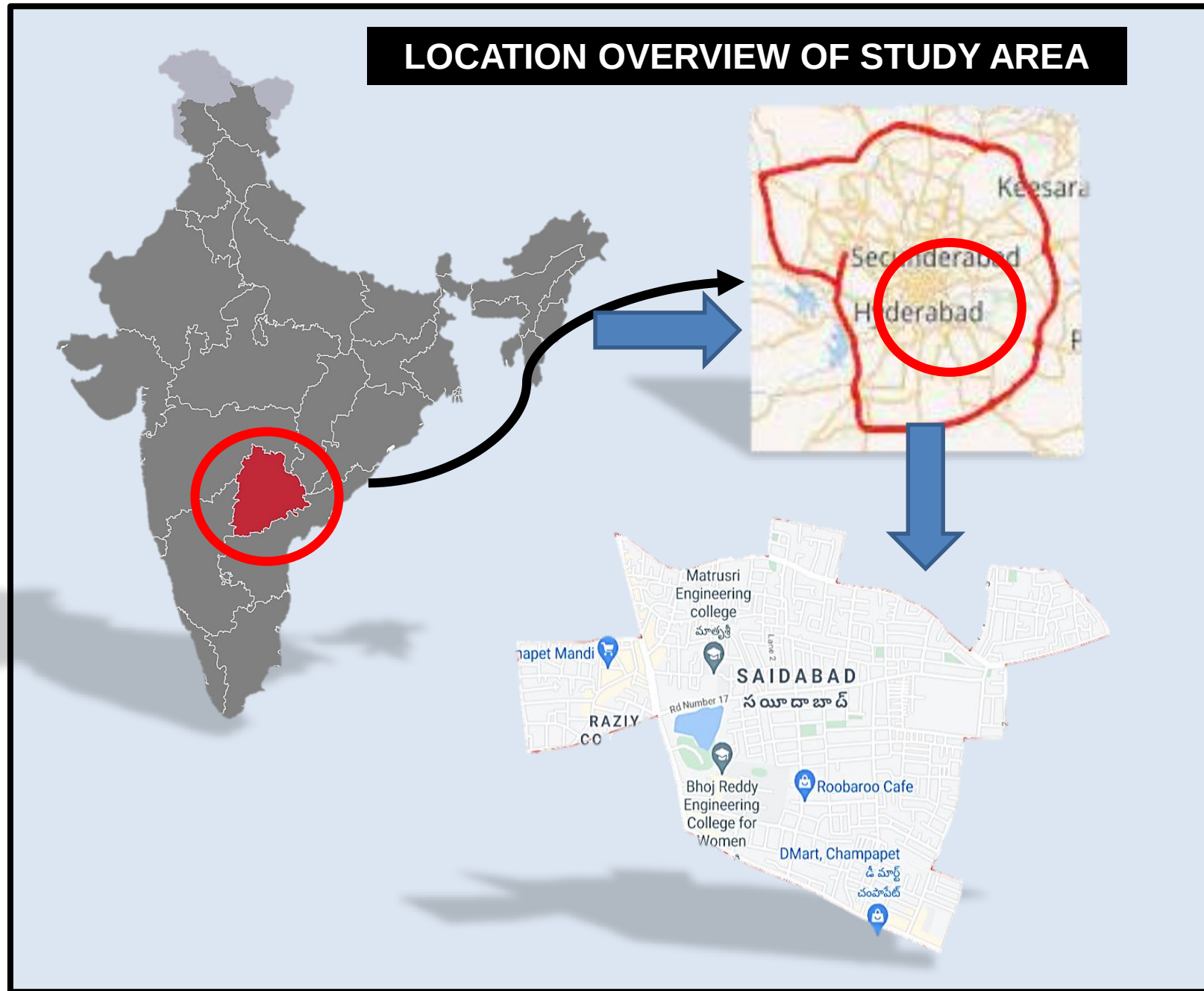
The three datasets with their parameters will be collated to prepare Comprehensive Green Space Map and Green Space Index for each ULB.

- Total of 141 ULBs in Telangana are evaluated.
- Some secondary data such as innovations, best practices, selection of tree species, landscape design shall be collected from ULBs to help identifying extra efforts by the ULBs in creating pleasing green environments.

National institute of urban management shall carry out the evaluation for green space index and furnish a report and ULBs will be awarded based on six different arenas of performances.

STUDY AREA PROFILE

LOCATION OVERVIEW OF STUDY AREA



As per the Population Census 2011 data, following are some quick facts about Saidabad Mandal.

	Total	Male	Female
Children (Age 0-6)	39,324	20,680	18,644
Literacy	84.66%	77.86%	72.05%
Scheduled Caste	20,336	10,400	9,936
Scheduled Tribe	5,809	3,068	2,741
Illiterate	86,335	39,244	47,091

SAIDABAD is a mandal falls under the GHMC(GREATER HYDERABAD MUNICIPAL COPORATION) boundary of Hyderabad the capital city of Telangana state. The city is one of the largest metropolitan of India with an area of 625.9 square kilometers (250 sq mi) with the population of 6,809,970 and further 7,749,334 living in the metropolitan area, making Hyderabad the fourth most populous city and the sixth-most populous urban agglomeration in India.

DEMOGRAPHIC PROFILE OF STUDY AREA

- **AREA** : 6.59sq KM
- **total population** : 345,722 as per the Census 2011.
- **Male population** : 177,222
- **Female population** : 168,500 In 2011 there were
- **Total 74,462 families** residing in Saidabad Mandal.
- **Average Sex Ratio** of Saidabad Mandal is 951.
- **The average literacy rate** in rural area is 84.7%
- **Sex ratio** : 951/1000 male
- **The total literacy rate** of Saidabad Mandal is 84.66%
- **male literacy rate** is 77.86%
- **female literacy rate** is 72.05%
- **Population Density:** 18098 people per km²

GHMC Ward No – 18, 19, 20, 21 Saidabad Mandal wards

Population
345,722

Literacy
84.66%

Sex Ratio
951

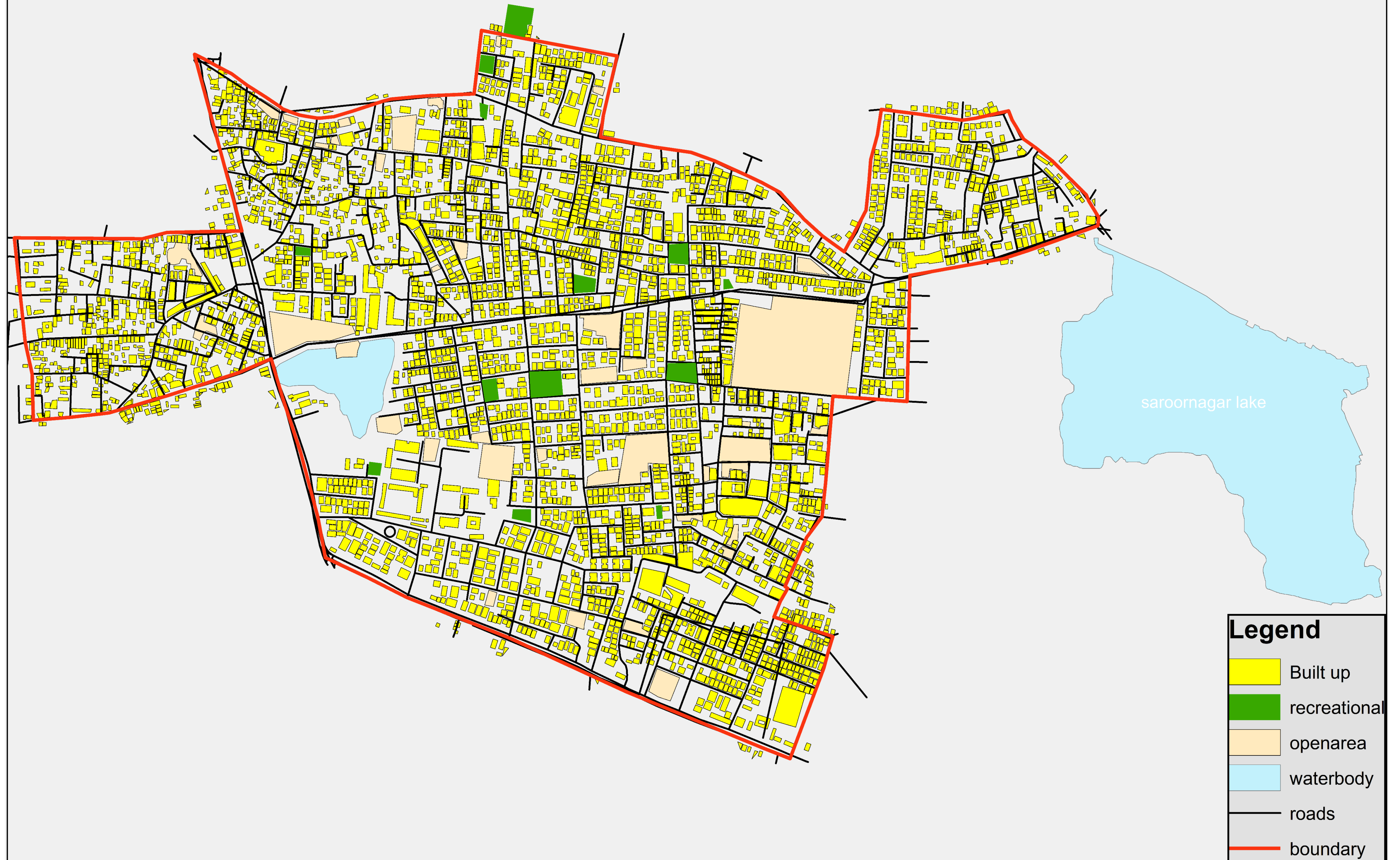
While the powers-that-be continue to twiddle their thumbs even in December, the city continues to grapple with the lack of quality green spaces. As per the figures of the Census 2011, the city has a population of 68.1 lakh and spans over 650 square kilometres. However, the figures, according to the GHMC website, show that the city has only 694.4 acres allocated for parks and open spaces, which is just 2.81 square kilometres. This means, on an average, there is only 2.7 square metres of open space per person in the city. As it turns out, the urban and regional development plans formulation and implementation (URDPFI) guidelines say that it is optimal to have at least 10-12 square metres per person. Even the national building code (NBC) prescribes that a person should be given at least three-square metres of organised green space.

“It is common knowledge that there is a lack of green spaces in the city. The parks that the GHMC looks to build, or has built, is not proportional to the population of the city. We only have colony parks, which are used for cultural reasons. They are not spaces where anyone can walk in or exercise,” Prof. B. Sri Nagesh, department of geography, Osmania University, told Deccan Chronicle.

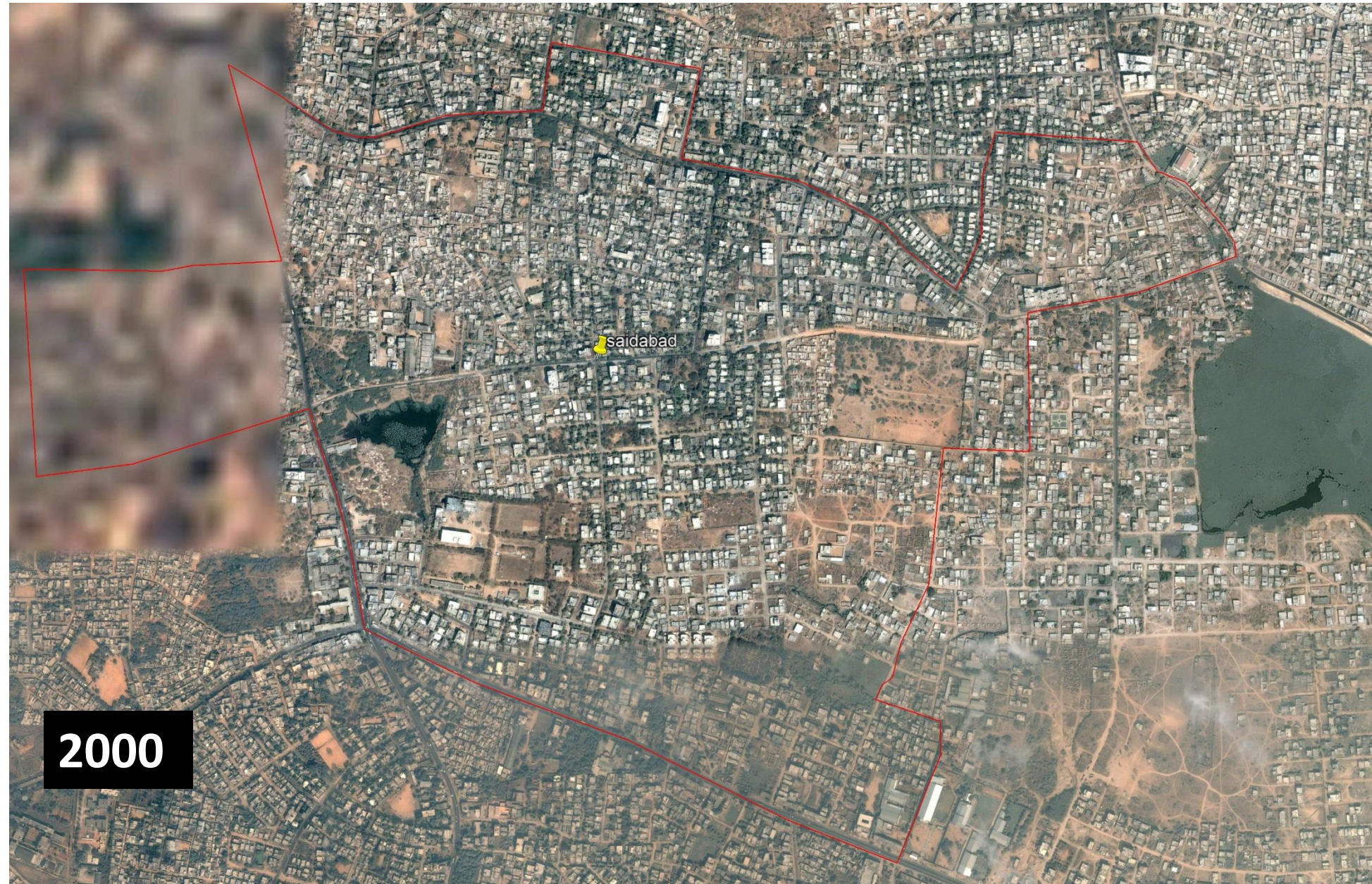
Source census of India <http://censusindia.gov.in/pca/pcadata/pca.html> <https://censusindia.gov.in/pca/pcadata/Houselisting-housing-AP.html>



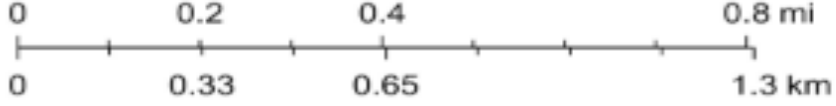
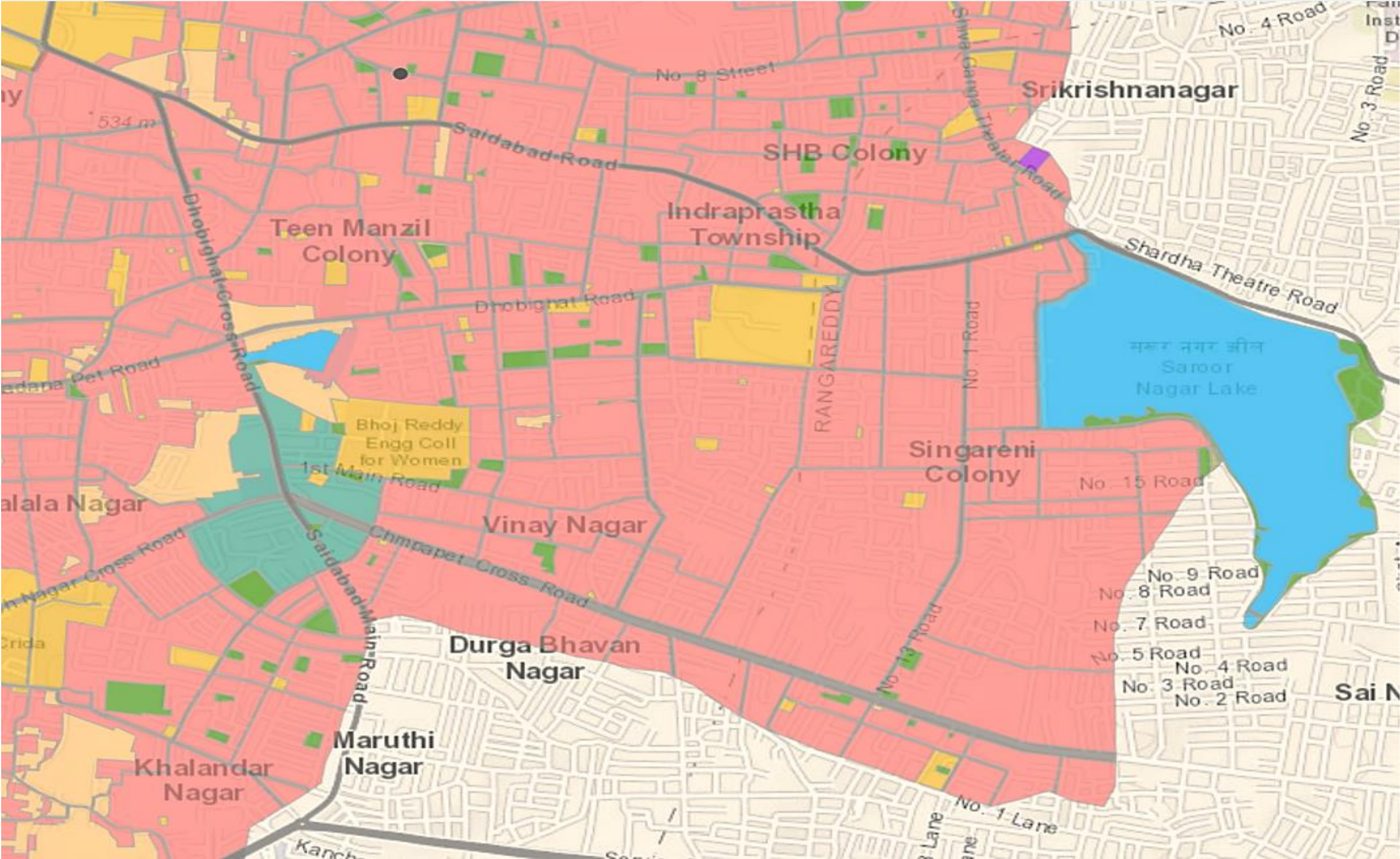
BASE MAP OF STUDY AREA SAIDABAD



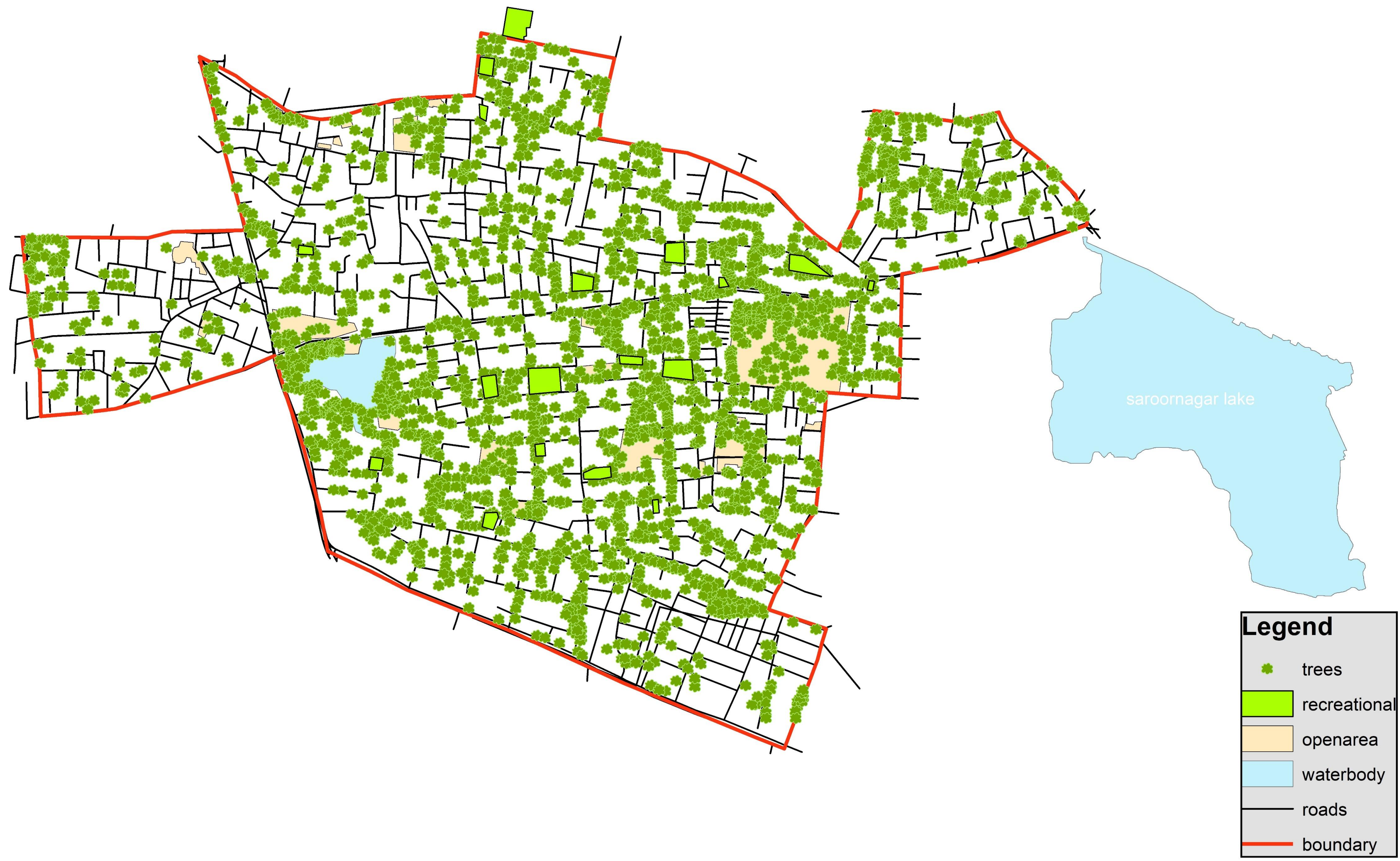
Timeline of the study area SAIDABAD 2000 – 2021 (10 years)



LANDUSE AND LAND COVER MAPS OF STUDY AREA SAIDABAD



GREEN COVER BASE MAP OF STUDY AREA SAIDABAD



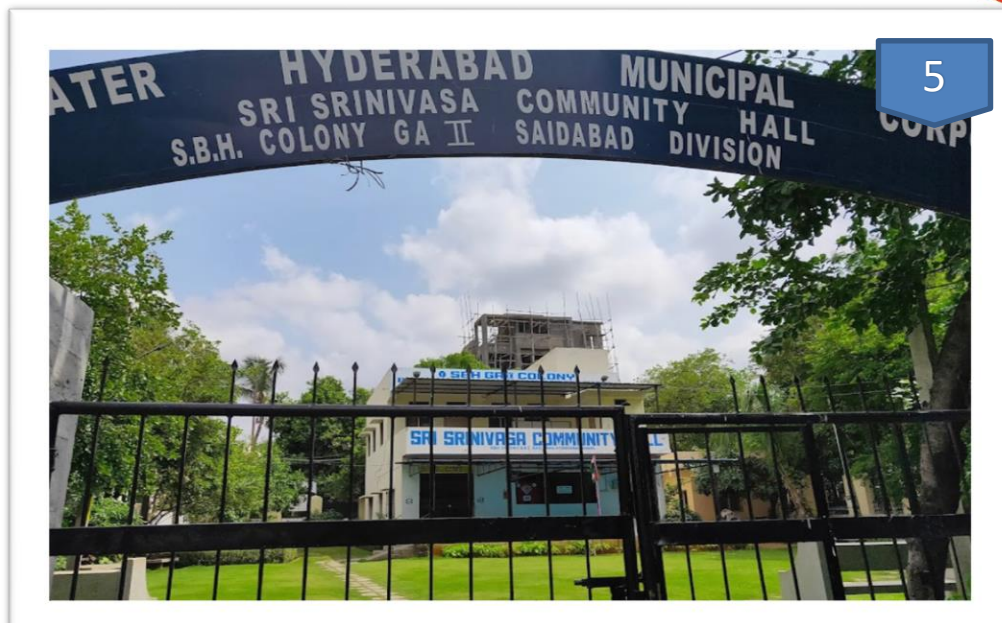
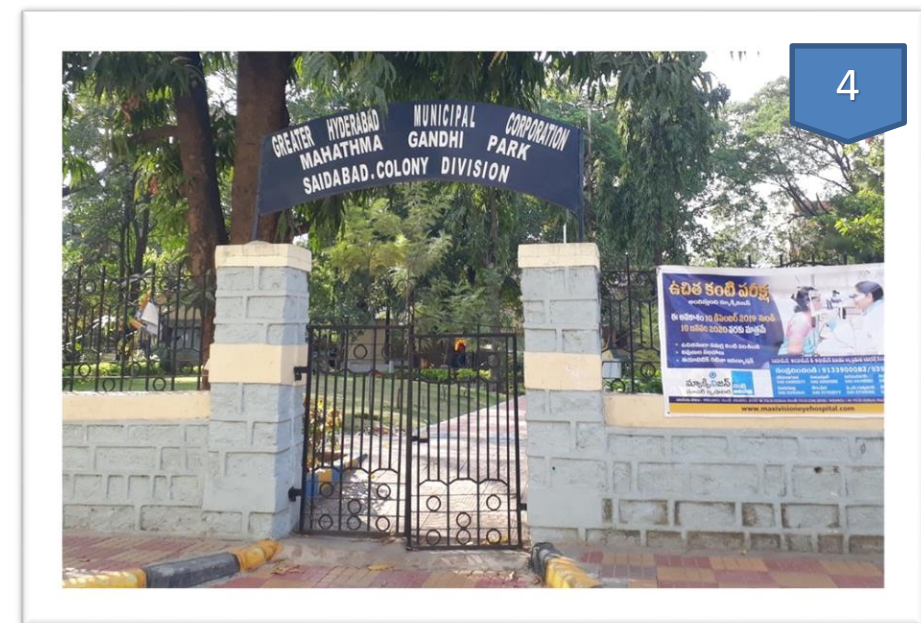
IDENTIFYING OF URBAN GREEN SPACES IN THE STUDY AREA :

INFERENCE: The total study area consists of few green spaces – parks, open spaces with total area of 5.46 acres out of 603.11 acres of study area.
The green spaces present in the study area are community park, tree park.

GS COMMUNITY PARKS in study area:

- 1. Subramanya Nagar park
- 2. Lakshmi Nagar municipal park
- 3.Municipal park
- 4.Mahatma Gandhi park
- 5.Sri Srinivasa community park

The other small parts of land covered with shrubs waste land.



STUDY AREA JUSTIFICATION:

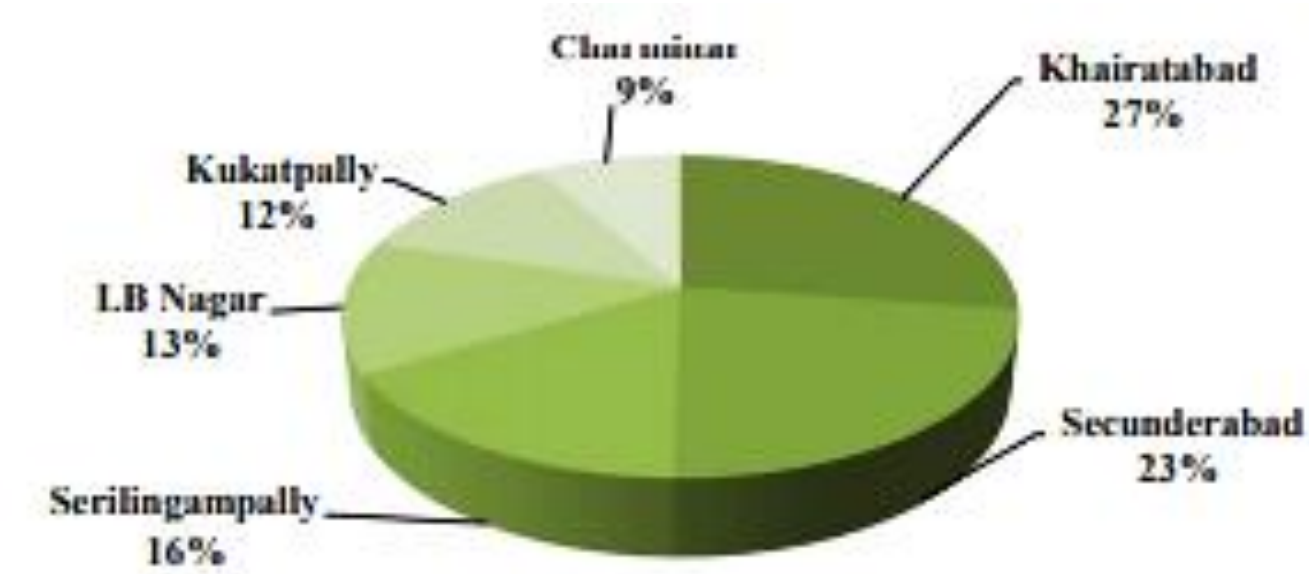
PERCENTAGE OF PARKS WITHIN ZONES OF GHMC

GREEN SPACE INDEX is a measure of green spaces available in particular neighborhood ,ward or city. The index will measure different parameters of green spaces .From the desk studies , parameters identifies to develop green space index for GHMC are -

S.no	ZONE	Population 2021 (in million)	Total ULB area (Sq. Kms)	Green cover (in Sq. Kms)	Green space provision per person (in sq.mts)
1	Charminar	1838679	98.694	28.65	0.12
2	Kukatpally	1206037	119.17	22.03	0.26
3	Khairatabad	1473880	78.063	17.25	0.49
4	Serilingampally	785196	136.258	21.09	0.55
5	LB Nagar	1168980	128.236	28.85	0.3
6	Secunderabad	1427228	65.41	16.01	0.43
TOTAL		7.9 million	625.831	133.87	

PARAMETER	PERCENTAGE	VALUE	QUALIFY CLASSES
Percentage of green in each cell	0-25	0.25	Low green
	25-50	0.50	Moderate green
	50-75	0.75	High green
	75-100	1	Very high

TABLE:PERCENTAGE OF GREEN



INFERENCE: GHMC has urban biodiversity resources such major parks , colony parks, tree parks and traffic islands with total area of 694.4 acres. There are about 159 traffic islands according to the GHMC. Khairtabad with 27% has more number of parks , next followed by secunderbad. Charminar has least percentage in coverage of parks in all the . And Saidabad Mandal comes under Charminar circle which has 9% of parks

- The green space index is shall be evaluated by collecting three different datasets
1. Green area map prepared using satellite imagery by GIS.
 2. Green space provision per person ,through projection of population data through census.
 3. Review of existing ongoing Urban greening projects through secondary survey by Urban biodiversity wing of GHMC.

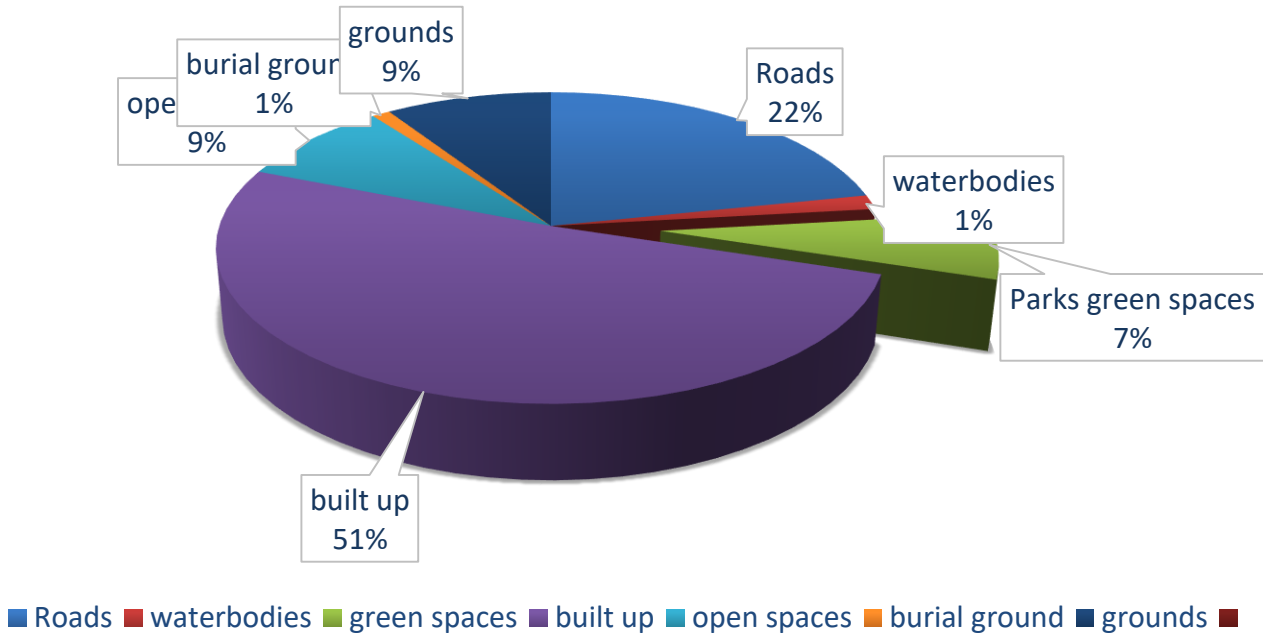
Percentage of green in each cell is computed based on green area classification .The high resolution satellite image has been generated for the study area GHMC.

ANALYSING CLASSIFICATIONS OF PUBLIC URBAN GREEN SPACE

Public urban green spaces represent a large group of spatial concepts, forms, and functions. They are parks, streets, gardens, commons, in-between spaces, sports fields, cemeteries, etc. It can be discerned from the names of 69 these forms that they are related to use, function, and form.

Based on the percentage, each cell has been classified into five green classes: very low, low, moderate, high and very high green quality on a scale of 0.2 – 1 where less than 20% of green in a cell has been categorized as very low quality green hence, given a value of 0.2. In the same manner, 0.4 (low green quality), 0.6 (moderate green quality) ,0.8 (high green quality) and 1 (very high green quality) value have been given to cells where percentage of green is 20-40%, 40-60% ,60- 80% and more than 80%, respectively

Fig :Landuse/Landcover of saidabad



GREEN SPACE INDEX FOR GHMC

GREEN SPACE INDEX is a measure of green spaces available in particular neighborhood ,ward or city. The index will measure different parameters of green spaces .From the desk studies , parameters identifies to develop green space index for GHMC are -

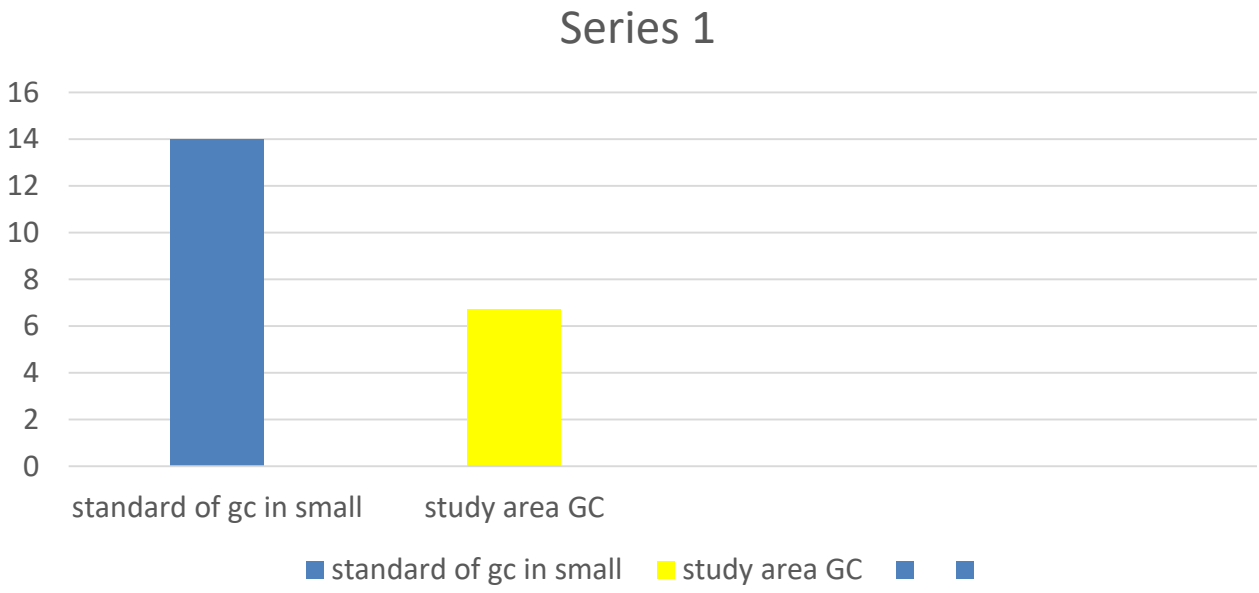
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Land Use Category	Percentage of developed area			
	Small	Medium	Large	Metro Cities
Residential	45 - 50	40 - 45	35 - 40	35 - 40
Commercial	2 - 3	3 - 4	4 - 5	4 - 5
Industrial	8 - 10	8 - 10	10 - 12	12 - 14
Public & Semi-Public	6 - 8	10 - 12	12 - 14	14 - 16
Recreational	12 - 14	18 - 20	18 - 20	20 - 25
Transport & Communication	10 - 12	12 - 14	12 - 14	15 - 18
Agriculture & Water bodies	Balance	Balance	Balance	Balance

Source: UDPFI Guidelines

The standard percentage of Recreation in small area which is considered as a ward in this analysis should have 12-14% . Which Is shown in blue color, a

S no.	Parameters	percentage
1	ROADS	21.7
2	Water bodies	1.5
3	Parks (GS)	6.71
4	Built up	51.2
5	Open spaces	8.7
6	Burial ground	1
7	grounds	9.2
Total percentage		100



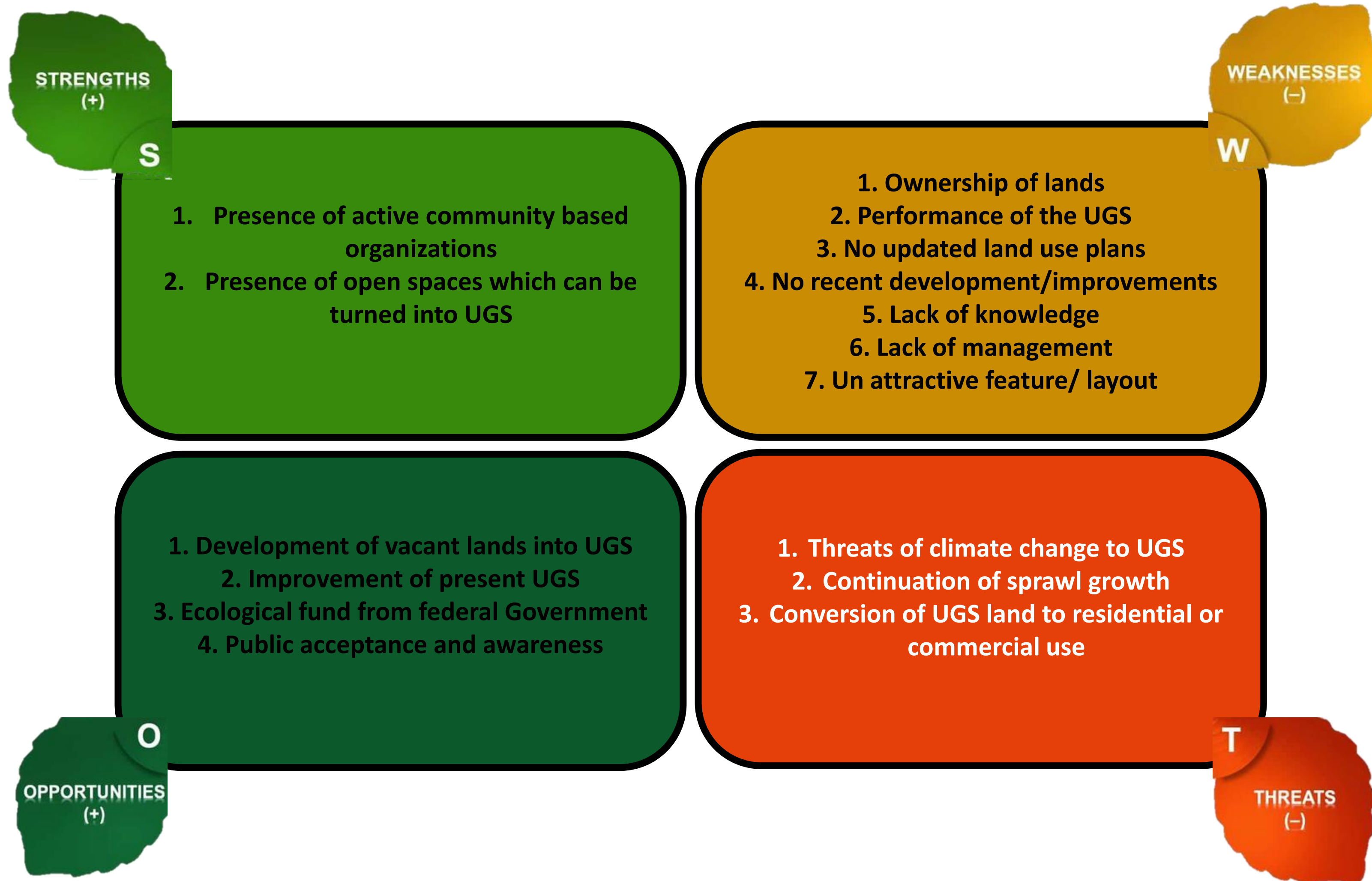
The standard percentage of Recreation in small area which is considered as a ward in this analysis should have 12-14% . Which Is shown in blue color, and the yellow bar is representing the percentage of recreation in study area which is Saidabad. Saidabad has 6.7 in total which is half of the standard.

S.No	Category	Planning Norms & Standards	
		Population/ unit (Approx)	Plot area (Ha)
1.	Neighborhood Park	10000	1.0
2.	Housing area park	5000	0.5
3.	Tot. Lot at Housing Cluster	350	0.0125

Table Planning Norms, standards for recreational Areas / Parks at neighborhood



SWOT ANALYSIS





Green Roofs:

Green Roofs is an innovative yet simple method to improve the existing green percentage of the area. If individuals start roof gardening or Roof plantations it will be Helpful bothways, increase the green percentage and also recreation for the family.

Green roofs are covered with growing media and vegetation that enable rainfall infiltration and evapotranspiration of stored water.



Many green infrastructure elements can be seamlessly integrated into parking lot designs. Permeable pavements can be installed in sections of a lot and rain gardens and bioswales can be included in medians and along the parking lot perimeter. Benefits include mitigating the urban heat island and a more walkable built environment.

RECOMMENDATION



Green streets are created by integrating green infrastructure elements into their design to store, infiltrate, and evapotranspire stormwater. Permeable pavement, bioswales, planter boxes, and trees are among the elements that can be woven into street design



LAND CONSERVATION

The water quality and flooding impacts of urban stormwater also can be addressed by protecting open spaces and sensitive natural areas within and adjacent to a city while providing recreational opportunities for city residents.

Assessment of vegetation in green spaces

Depending on the locations, the plant species, names of the main plants grown in major cities of India according to the Gujarat Forest Department report

Table 2. Green spaces and plant species grown in some major cities of India.

City	Green Space (%)	Major Plant Species
New Delhi	11.90	Banyan tree, Peepal tree, Bael, Jamun, Ber, Arjun
Mumbai	6.20	Bhendi tree, Banyan trees, Tamarind tree, Coconut palm, Paral tree, Padauk trees, Mahogany tree, Cajuput tree, Baobab tree, Star Apple, Baobab tree, Peltophorum tree, Gulmohar tree
Chennai	7.50	Sansiveria, Dieffenbachia, Dracaena, Spider plant, Earth star, Money plant, Pothos, Syngonium
Hyderabad	1.66	Bougainvillea, Neerium, Adeneum, Lantana, Euphorbia russelia
Bengaluru	2.96	Australian wattle, Butterfly tree, Red silk-cotton tree, Popcorn bush cedar, Coconut palm, Gulmohur, Indian cork tree, Indian elm, Teak, Silver oak, Orange champak
Kolkata	7.30	Neem, Peepal, Banyan, Radhachura, Krishnachura, Tamarind, Coconut, Betelnut tree
Pune	1.40	Cadamb, Tetu, Awala, Chandan, Tamhan, Muchkund, Cadamb, Kanchan, Putranjiva, Semla Kanchan, Kapila, Murudsheng Ritha, Undi
Panaji	10.00	Coconut, Teak, Sal, Cashew, Mango, Jackfruit tree

The table , Above is the types of plants that are suitable for the cities. Hyderabad city can accommodate tress like Neem , Lantana , etc.



Planter boxes are urban rain gardens with vertical walls and either open or closed bottoms. They collect and absorb runoff from sidewalks, parking lots, and streets and are ideal for space-limited sites in dense urban areas and as a streetscaping element

